Integrated Working Edition

This merges the completed conceptual subtopic guides. Concept coverage is mature, but the ExamSIDE corpus audit is still closing question-level gaps; see the separate coverage audit report.

CONTENTS

THE COUNTING ENGINE

Real situations -> visual models -> product rule -> factorial -> ordered selection

GOLD		SILVER		BRONZE
7	x	6	x	5

Essential Question

How can one counting engine explain medals, codes, lifts, matrices and functions?

THIS SUBTOPIC BUILDS

1. Sequential-choice thinking
2. Repetition vs shrinking pools
3. n! and nPr as compressed products

LESSON CYCLE

Experience -> Notice -> Model -> Name

Guided Practice -> Misconception Repair

JEE Transfer -> Check -> Synthesis

HOW THIS SUBTOPIC WORKS

One dominant instructional job per page

ST01

Concept spine

- C1 — **Order changes outcomes** — swap test
- C2 — **Product rule** — sequential choices
- C3 — **Repetition** — same pool vs shrinking pool
- C4 — **Factorial and nPr** — compression
- C5 — **Method control** — slots vs nPr

SCOPE

Boundary

Blocks, gaps, repeated objects, number-formation restrictions, dictionary rank and circular permutations are deliberately deferred.

Release standard for every core concept

1. **Real-life entry**
2. **Meaning before notation**
3. **Faithful concept helper**
4. **Worked teacher model**
5. **Misconception + repair**
6. **JEE trigger wording**
7. **Disguised transfer**

READABILITY

Typography gate

Content text is larger in v5. Dense pages are split instead of shrinking type to make everything fit.

START WITH EXPERIENCES STUDENTS ALREADY UNDERSTAND

Entry A: labelled outcomes and repeated slots

ENTRY A

REAL LIFE

Podium medals

Ten finalists compete for Gold, Silver and Bronze. If A wins Gold and B wins Silver, swapping A and B changes the result because the medal labels are different.

GOLD		SILVER		BRONZE
10	x	9	x	8

REAL LIFE

Access code

A 4-character access code uses six symbols. When reuse is allowed, filling one position does not remove that symbol from the pool for later positions.

1		2		3		4
6	x	6	x	6	x	6

NOTICE

Compare before abstracting

The stories differ, but both are built from labelled positions + available choices. The medal problem has a shrinking pool; the code problem keeps the same pool.

ENTRY B

REAL LIFE

Lift exits

Three labelled passengers leave at three different permitted floors. Choosing only the set of floors is incomplete: the final outcome also records which passenger leaves at which floor.

P1		P2		P3
7	x	6	x	5

REAL LIFE

Photo line-up

Five distinct people stand in five labelled positions for a photograph. Every person is used exactly once, so each placement removes one person from the remaining pool.

1		2		3		4		5
5	x	4	x	3	x	2	x	1

SYNTHESIS

The counting engine

Medals, codes, lift exits and line-ups are not four unrelated tricks. They are four versions of the same question: what are the labelled slots, and what remains available for each slot?

C1 | ORDER: WHEN DOES A NEW OUTCOME APPEAR?

Use the swap test before any formula

C1

Compare two tasks using the same students

REAL LIFE

Assign two offices

From six students, assign President and Secretary. Swapping the two selected students changes the role assignment, so it creates a new outcome.

A -> President, B -> Secretary != B -> President, A -> Secretary

REAL LIFE

Choose two representatives

Choose two students only to attend a meeting. The group {A,B} is the same selected pair as {B,A}; swapping creates no new outcome.

$$\{A,B\} = \{B,A\}$$

Concept helper: SWAP TEST

HELPER

Swap before formula

Imagine two chosen objects exchange roles or positions. If the recorded outcome changes, order matters. If the outcome is unchanged, you are only selecting.

MISCONCEPTION

ORDER IGNORED

Tempting thought: “The same people are chosen, so the count should not depend on order.”

Why it fails: The outcome includes labelled roles. The same people in different roles are different outcomes.

Repair: Write the role labels explicitly, then perform the swap test.

JEE TRIGGER

Language cue

Words such as rank, role, position, seat, digit-position, image, destination often create labelled slots.

C2 | PRODUCT RULE: COUNT A PROCESS, NOT A LIST

C2

Teacher model: student council roles

Six students are available for President, Secretary and Treasurer. Nobody may hold two roles.

P		S		T
6	x	5	x	4

$$6 \times 5 \times 4$$

WHY MULTIPLY **Sequential choices**

Each President choice can be followed by each legal Secretary choice, and each of those by each legal Treasurer choice. All three stages happen: first THEN second THEN third.

CHECK **Student action**

Nine students are available for three distinct offices. Write only the slot counts and explain why they shrink.

Misconception repair

MISCONCEPTION **ADD INSTEAD OF MULTIPLY**

Tempting thought: “There are three visible counts, so add: 6 + 5 + 4.”

Why it fails: Addition counts alternatives such as case A OR case B. Here all stages occur in the same outcome.

Repair: Ask “OR?” versus “THEN?”. Sequential stages multiply.

CONCEPT **Tree -> product**

The Product Rule is the branch count of a decision tree written compactly. You do not need to draw every branch once the repeated structure is visible.

JEE TRIGGER **Language cue**

Look for ordered roles, seats, positions, stages, destinations or successive choices. Translate the sentence into slots before touching notation.

CONCEPT HELPER TOOLKIT

Use the smallest representation that repairs recognition

HELP

H1 - Question-to-slot translator

H1 **Recognition cue**

Ask two questions: What are the labelled positions? and what can enter each position? This changes what you notice without supplying the setup.

H2 - Choice-pool visual

H2 **Visual model**

Write the available pool beside each slot. Cross out a used object only when the rules forbid reuse. This makes repetition assumptions visible.

H3 **Setup only**

Write only the product or case skeleton, such as 7 x 6 x 5, and stop before arithmetic. H3 rescues setup without giving the answer.

Worked helper example: event badge

A badge contains a letter, then a digit, then a letter. Five letters and four digits are available; reuse is allowed.

LETTER		DIGIT		LETTER
5	x	4	x	5

NOTICE **Availability is local**

The number of choices can change from slot to slot even when reuse is

MISCONCEPTION **ALL SLOTS SAME SIZE**

Tempting thought: “If repetition is allowed, every factor must be the same.”

Why it fails: Repetition controls reuse; it does not make different slot-types identical.

Repair: Count each slot from its own rule. Powers are only a special case.

C3 | REPETITION: DOES A USED OBJECT STAY AVAILABLE?

Same positions, different pool rule

C3

Real-life comparison: access codes

REUSE ALLOWED

1		2		3		4	
6	x	6	x	6	x	6	6^4

NO REUSE

1		2		3		4	
6	x	5	x	4	x	3	6 x 5 x 4 x 3

PREDICT — **Reason before arithmetic**

Before calculating, which version must have more outcomes? Explain using the size of the later choice pools.

The one-question pool test

HELPER — **Pool rule**

After filling a slot, ask: Is the chosen object still legally available? If yes, the pool need not shrink. If no, remove it before the next slot.

MISCONCEPTION — **REPETITION ACCIDENTALLY FORBIDDEN**

Tempting thought: “Different positions must contain different symbols.”

Why it fails: Distinct positions and distinct symbols are different statements. Labelled slots alone do not forbid reuse.

Repair: Read the reuse rule independently from the position labels.

JEE TRIGGER — **Language cue**

Look for may repeat, repetition allowed, without repetition, distinct digits/letters, used at most once.

C4 | FACTORIAL AND nPr: NAME THE SHRINKING PRODUCT

Notation comes after the model

C4

Full arrangement: photo line-up

REAL LIFE

Five-person photo

Five distinct people stand in five labelled positions. Every person is used once, so the pool shrinks until no person remains.

1		2		3		4		5
5	x	4	x	3	x	2	x	1

MEANING

Use every object

Factorial is only a compact name for a full shrinking product.

Partial arrangement: podium

REAL LIFE

Three medals from seven finalists

Only three labelled positions are filled, so the shrinking product stops after three factors.

GOLD		SILVER		BRONZE
7	x	6	x	5

MEANING

Use only r positions

nPr is the same shrinking-pool idea stopped after r labelled slots.

C5 | METHOD CONTROL: SLOTS OR nPr?

Choose the representation that exposes restrictions

C5

Same answer, different visibility

Eight students are available for Captain, Vice-Captain and Treasurer.

SLOTS

C	VC	T
8 ×	7 ×	6

Best when a restriction attaches to a particular position.

nPr

$$^{8}P_{3}$$

Best when the ordered slots are structurally identical and unrestricted.

Decision cues

SLOTS FIRST

Restrictions are visible

Use slots when the question mentions first/last/middle, a fixed place, a forbidden object, zero-leading, or another slot-specific rule.

nPr IS CLEAN

Compact form

Use nPr when r distinct roles or destinations are filled from n distinct objects with no extra positional complication.

MISCONCEPTION

FORMULA-FIRST DRIFT

Tempting thought: “Writing nPr immediately is faster.”

Why it fails: Notation can hide a special position or case split before you have modelled it.

Repair: Build the slots first, then compress only if the structure stays uniform.

C6A

Lift exits: people -> distinct destinations

REAL LIFE

Three passengers

Three labelled passengers leave at three different permitted floors. Treat each passenger as a labelled source and each floor as a destination.

P1		P2		P3
7	x	6	x	5

Permutation matrix: rows -> unused columns

CONCEPT TRANSFER

In a 5 x 5 zero-one matrix with every row sum and column sum equal to 1, each labelled row chooses exactly one column, and no column can be reused.

0	1	0	0	0
0	0	0	1	0
1	0	0	0	0
0	0	0	0	1
0	0	1	0	0

5 x 4 x 3 x 2 x 1

MISCONCEPTION

Missing assignment stage

Choosing 3 floors with 7C3 records only the destination set. It does not record which passenger leaves at which floor.

TRANSLATOR

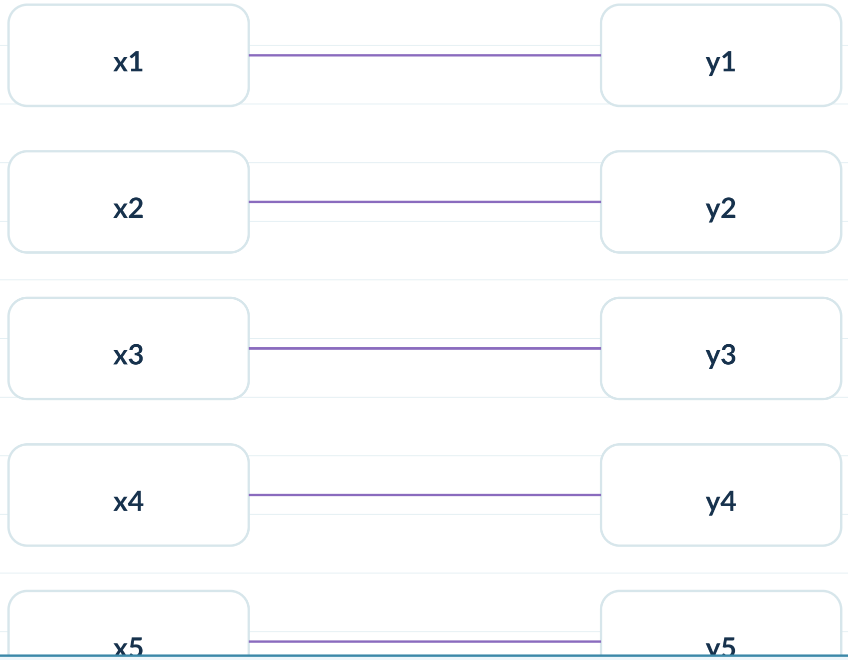
Assignment lens

Ask: what is being assigned to what? If labelled sources require distinct destinations, a hidden permutation may be present.

C6B

One-to-one function as a destination assignment

Suppose a domain has five labelled inputs and the available range has five distinct values. A one-to-one function must send different inputs to different outputs.



MODEL

Injection -> permutation

The first input has 5 output choices, the next 4, then 3, 2, 1. That is a factorial structure.

Concept helper: assignment sentence

HELPER

Surface wording -> structure

Rewrite the question as one sentence: “Each labelled ______ is assigned to one distinct ______.” Fill the blanks with the objects in the problem.

MISCONCEPTION

COUNTING ARBITRARY FUNCTIONS

Tempting thought: “Five inputs and five outputs gives 5 choices for every input, so 5^5.”

Why it fails: That allows two inputs to share an output, violating one-to-one.

Repair: After an output is used, remove it from the available pool

TRANSFER

One structure, many surfaces

The same lens works for seats, destinations, matrix columns, images of a function, ranks and labelled positions.

H1 -> H2 -> H3 HELPERS

Help enough to restore the model, not enough to steal the solution

H-LADDER

H1 - recognition cue

H1 **Recognition**

A short question that changes what you notice. Example: “Does the pool shrink after a choice?” No numerical setup yet.

H2 - visual / concept helper

H2 **Representation**

A slot row, pool sketch, mapping, tree or matrix that makes the structure visible. It should not finish the count.

H3 **Setup only**

Provide the product or case skeleton, then stop before arithmetic or the final answer.

Example ladder: lift exits

H1 **Recognition**

Choosing the set of floors is incomplete. What else is recorded in the outcome?

H2

P1		P2		P3
7	x	6	x	5

H3 **Setup**

Structural skeleton: 7 x 6 x 5. Do not evaluate until you can explain every factor.

MISCONCEPTION CLINIC I

Order and repetition errors

M1

MISCONCEPTION

ORDER IGNORED

Tempting thought: “Use 8C3 for Gold, Silver and Bronze.”

Why it fails: 8C3 only records who is in the winner set; it does not record which winner receives which medal.

Repair: Apply the swap test. If swapping Gold and Silver changes the outcome, order matters.

MISCONCEPTION

REPETITION ACCIDENTALLY ALLOWED

Tempting thought: “Use 6^4 even when symbols may not repeat.”

Why it fails: A power silently restores the same six choices at every position, including symbols already used.

Repair: Cross out a used symbol whenever reuse is forbidden; the pool must shrink.

MISCONCEPTION

REPETITION ACCIDENTALLY FORBIDDEN

Tempting thought: “Use 6 x 5 x 4 x 3 when reuse is allowed.”

Why it fails: Students confuse different positions with different symbols. The slots are distinct, but a symbol may legally reappear.

Repair: Read the repetition rule separately. If reuse is allowed, do not remove the chosen symbol.

SELF-CHECK

Explain the error, not just the answer

For each misconception above, write one sentence naming the hidden assumption that makes the wrong expression look plausible.

MISCONCEPTION CLINIC II

Assignment, addition and formula-first errors

M2

MISCONCEPTION

MISSING ASSIGNMENT STAGE

Tempting thought: “Use 7C3 for three passengers leaving at different floors.”

Why it fails: The set of floors is only one layer of the outcome. The passengers are labelled and must also be assigned to those floors.

Repair: Model passenger slots directly, or choose floors and then arrange passengers among them.

MISCONCEPTION

ADD INSTEAD OF MULTIPLY

Tempting thought: “Use 6 + 5 + 4 for three roles.”

Why it fails: The numbers represent stages that all occur in the same outcome, not mutually exclusive alternatives.

Repair: Use the OR-versus-THEN test. OR suggests addition; THEN suggests multiplication.

MISCONCEPTION

FORMULA-FIRST DRIFT

Tempting thought: “Write nPr before reading the restrictions.”

Why it fails: Notation can hide a special first position, forbidden object or case split.

Repair: Build the slots and their legal pools first. Compress only after the restrictions are visible.

REPAIR HABIT

Diagnose structurally

When an answer is wrong, do not only correct the arithmetic. Name the model assumption that failed: order, reuse, assignment, OR/THEN, or restriction visibility.

GUIDED PRACTICE: THE SCAFFOLD FADES

Model -> complete -> explain -> transfer

GP

A. Complete the structure

MODEL

5 runners compete for first and second.

5 x 4 = ______

COMPLETE

A 4-symbol code uses A-E without repetition.

5 x ____ x ____ x ____

COMPLETE

Choose and order 3 speakers from 9 students.

EXPLAIN

Reason in words

Choose one item and explain why the pool stays the same or shrinks.

B. Transfer without a method label

TRY 1

A theatre has 7 labelled VIP seats and 4 invited guests. Seat all 4 guests.

TRY 2

A locker PIN has 5 positions and 4 symbols; reuse is allowed.

TRY 3

Eight finalists receive Gold, Silver and Bronze.

MISCONCEPTION CHECK

Learn from the wrong model

For one item, write the most tempting wrong expression and name the assumption it makes.

JEE TRANSFER: RECOGNISE THE ENGINE UNDER EXAM WORDING

Representative ExamSIDE fingerprints

PYQ

Q1 | 2018

C3

Wording: n-digit numbers using only 2,5,7

Structural fingerprint: equal repeated slots -> power

OPEN EXAMSIDE

Q2 | 2026

C4/C6

Wording: three persons leave a lift at different floors

Structural fingerprint: distinct people -> distinct destinations

OPEN EXAMSIDE

Q3 | 2023

C6

Wording: 5x5 zero-one matrix; row/column sums 1

Structural fingerprint: each row chooses one unused column

OPEN EXAMSIDE

Q4 | 2024

C6

Wording: one-to-one functions after a 5-value range

Structural fingerprint: labelled inputs -> distinct outputs

OPEN EXAMSIDE

CHECK + MASTERY DASHBOARD

Can you recognise, represent, explain and transfer?

MASTER

Retrieval check

1. Five distinct books are placed in a row.
2. Three medals are awarded to 10 finalists.
3. A 4-character code uses six symbols with repetition allowed.
4. Seven people occupy four labelled chairs.
5. Explain why 7C3 undercounts the lift-exit problem.

ANSWERS **Check after attempting**

1. 5!   2. 10P3   3. 6^4   4. 7P4   5. it selects floors but does not assign passengers.

Mastery self-check

1. I can use the swap test.
2. I can draw slots before notation.
3. I can decide whether a used object remains available.
4. I can explain n! and nPr as shrinking products.
5. I can identify the misconception in a wrong solution.
6. I can translate lift, matrix and function wording into assignments.
7. I can use H1/H2/H3 without jumping straight to H3.

EXIT CONDITION **Ready for next subtopic**

Move to ST02 only when a new restriction can be added without reteaching the product-rule engine.

A1

Same objects, different outcome definition

PERMUTATION

Objects occupy distinct roles, positions or destinations. Swapping two assigned objects creates a new outcome.

Gold=A, Silver=B != Gold=B, Silver=A

COMBINATION

Only membership of the chosen set matters. Reordering the same members does not create a new selection.

{A,B} = {B,A}

Hybrid: choose, then arrange

REAL LIFE

Lift example

Choose three permitted floors, then assign three distinct passengers one per floor. Selection is one stage; ordered assignment is another.

EQUIVALENT

Two routes, one count

You can count the final ordered assignment directly as nPr, or as “choose destinations” then “arrange labelled passengers.” Both count the same final

STOP POINT

Preliminary only

Do not learn nCr formulas here. The only guardrail needed for ST01 is: swap changes outcome -> permutation thinking; swap does not -> selection thinking.

APPENDIX A | CLASSIFICATION DIAGNOSTIC

Write P, C or Hybrid; compute nothing

A2

Classify the outcome structure

1. Choose 3 students for a committee. P / C / H
2. Assign President, Secretary and Treasurer from 8 students. P / C / H
3. Choose 4 books, then arrange them on a shelf. P / C / H
4. Choose 3 floors and assign 3 passengers one per floor. P / C / H
5. Select 5 digits, then form a 5-digit number. P / C / H

ANSWER **Check classification**

1 C 2 P 3 H 4 H 5 H 6 C

Guardrails

MISCONCEPTION **WORD "CHOOSE" = COMBINATION**

Tempting thought: "If the question says choose, it must be a combination."

Why it fails: Exam wording is not the mathematical object. The chosen objects may later occupy labelled roles or positions.

Repair: Ask what the final outcome records, then perform the swap test.

MISCONCEPTION **WORD "ARRANGE" = ONE FORMULA**

Tempting thought: "If the question says arrange, write nPr immediately."

Why it fails: Extra restrictions may require special slots, blocks, gaps or cases.

Repair: ST01 gives the engine; later subtopics teach restriction machinery.

REPEATED OBJECTS

When swapping identical copies creates no new arrangement

R	R	B	B	B

Essential Question

Why does n! overcount when some objects are indistinguishable?

THIS SUBTOPIC BUILDS

1. Identity-swap reasoning
2. Multiset permutation formula
3. Position-selection equivalence

JEE TRANSFER

Repeated symbols -> word counts

Choose pattern -> arrange multiset

Repeated-digit positional sums

HOW THIS SUBTOPIC WORKS

One idea: identical swaps must not be counted again

ST02

Concept spine

- C1 **Identical-copy swap test** — what counts as new?
- C2 **Why n! overcounts** — temporarily label copies
- C3 **Multiset formula** — divide invisible swaps
- C4 **Position-selection view** — choose slots for types
- C5 **Choose then arrange** — hybrid construction

SCOPE **Boundary**

Blocks, gaps, dictionary rank, circular permutations and digit-divisibility restrictions are deferred. Repeated objects may appear inside those later subtopics as secondary machinery.

Pedagogy layers used here

1. Real-life patterns
2. Meaning before formula
3. Identity-swap visual
4. Teacher model
5. Guided completion
6. Misconception repair
7. JEE trigger language
8. Disguised transfer
9. Retrieval + audit

START WITH A VISIBLE PATTERN, NOT A FORMULA

ENTRY A

REAL LIFE

Archive shelf

A clerk places 3 identical red folders and 2 identical blue folders in a row. Swapping two red folders is physically possible, but the visible shelf does not change.

R	R	R	B	B

REAL LIFE

Signal lights

A monitoring display records a 6-slot history with exactly four green signals and two amber signals. Only the pattern of colors matters; individual green signals have no identity.

G	G	G	G	A	A

NOTICE

The structural difference

In ST01 every object was distinct. Here some copies are indistinguishable. A swap between identical copies creates no new recorded outcome. That single change is why ordinary factorial counting overcounts.

THE SAME STRUCTURE HIDES INSIDE WORDS AND SEQUENCES

Entry B: repeated symbols are a multiset

ENTRY B

REAL LIFE

Repeated letters

Suppose a label is made from the letters A, A, B, C. If we secretly call the A copies A1 and A2, two labelled arrangements can collapse to the same visible word when the subscripts are erased.

A_1 A_2 B C

REAL LIFE

Binary result string

A test record has 8 positions with exactly 5 correct marks C and 3 wrong marks W. The question is not which C is which; it is which positions contain C.

C C C C C W W W

SYNTHESIS

One engine, many surfaces

Words, color-patterns and repeated-result strings all ask for arrangements of a multiset: some symbols occur more than once and identical copies are not separately observable.

C1 | THE IDENTICAL-COPY SWAP TEST

Ask whether a swap changes the visible outcome

C1

Compare two arrangements

A	A	B	C
1	2		

A1 A2 B C -> erase labels -> A A B C

A	A	B	C
2	1		

A2 A1 B C -> erase labels -> A A B C

CORE

What overcount means

The two labelled arrangements are different, but after the artificial labels are erased they become the same visible arrangement. Therefore n! counts each visible arrangement multiple times.

Misconception repair

MISCONCEPTION

EVERY SWAP IS NEW

Tempting thought: “Any two objects swapping positions must create a new permutation.”

Why it fails: That is true only when the objects are distinguishable in the recorded outcome. Identical copies leave the visible arrangement unchanged.

Repair: Before using n!, perform the identity-swap test: can the observer tell the two copies apart?

JEE TRIGGER

Language cue

Repeated letters, repeated digits, exact counts of symbols, identical colored objects, or phrases such as “used twice/thrice” should trigger an indistinguishability check.

C2

Teacher model: A, A, B, C

Pretend the two A copies are different: A1 and A2. Then all four objects are distinct, so there are 4! labelled arrangements.

A 1	A 2	B	C

labelled count = 4!

NOTICE

Invisible swaps

For every visible word, the two fake A-labels can be swapped in 2! ways without changing the visible result. So the labelled total counts each visible arrangement exactly 2! times.

visible count = 4! / 2!

Generalize the correction

If n objects contain groups of identical copies of sizes a, b, c, ... , then each visible arrangement was overcounted by a!b!c!... artificial labelings.

CONCEPT

Divide only for indistinguishable copies

Multiset permutation:

$$n! / (a!\, b!\, c!\, \ldots)$$

where a+b+c+...=n.

MISCONCEPTION

DIVIDE BY NUMBER OF TYPES

Tempting thought: “There are three kinds of symbols, so divide by 3!.”

Why it fails: Overcount comes from permutations within each identical group, not from the number of symbol-types.

Repair: Write the multiplicity of each repeated type and divide by each multiplicity factorial.

C3 | MULTISET PERMUTATIONS IN ONE LOOK

Count multiplicities before calculating

C3

Example 1: BALLOON

BALLOON has 7 letters. L appears twice and O appears twice; the other letters appear once.

B	A	L	L	O	O	N

$$7! / (2!\,2!)$$

TEACHER THINKING **Decision sequence**

1. Count all positions.
2. Record multiplicities.
3. Divide by swaps that are invisible.
4. Do not divide for letters that occur once.

Example 2: 10-term sequence

A sequence contains exactly five 1s, three 2s and therefore two 0s. The ten slots are distinct; only the repeated symbols are indistinguishable.

1	1	1	1	1	2	2	2	0	0

$$10! / (5!\,3!\,2!)$$

JEE FINGERPRINT **Exam transfer**

A recent JEE Main question uses exactly this structure: ten terms from {0,1,2} with prescribed counts of 1s and 2s. This is a direct multiset-permutation trigger.

C4 | A SECOND REPRESENTATION: CHOOSE POSITIONS

Sometimes combinations show the same multiset count more clearly

C4

Same 10-slot sequence, different view

Instead of permuting ten repeated symbols at once, choose where each

MODEL

Position selection

Choose 5 of the 10 positions for the 1s.

Then choose 3 of the remaining 5 positions for the 2s.

The last 2 positions are automatically 0s.

$$C(10,5) \times C(5,3)$$

Why both methods agree

$$C(10,5)\ C(5,3) = 10! / (5!\ 3!\ 2!)$$

CONCEPT

Two representations, one count

The factorial-division view counts visible arrangements directly. The position-selection view constructs the same arrangement by assigning slots to symbol-types.

METHOD CONTROL

Choose clarity

Use the representation that exposes the condition. Exact symbol counts often favor position-selection; plain repeated-letter arrangements often favor factorial division.

C5

Teacher model: SYLLABUS-type structure

Suppose the available letters include two S copies, two L copies, and

MODEL

Choose -> choose -> arrange

Stage 1 - choose the repeated type.
Only letters having at least two copies are eligible.

Stage 2 - choose two other distinct letters.

Stage 3 - arrange the multiset.
Pattern: A, A, B, C -> 4!/2!.

JEE TRIGGER

Inventory before arrangement

Phrases such as "two alike and two distinct", "one digit repeated exactly twice", or "two digits each repeated twice" signal that the multiplicity pattern itself must first be chosen.

Misconception repair

MISCONCEPTION

ARRANGE BEFORE INVENTORY

Tempting thought: "Start with 4! and then correct somehow for repetition."

Why it fails: Before arranging, the question may not yet have determined which symbols are present or which symbol is repeated.

Repair: Write an inventory pattern first: e.g. 2+1+1. Then choose the symbol-types, then arrange that multiset.

HELPER

Multiplicity template

H2 visual: write a multiplicity template such as [2,1,1] or [2,2,1,1,1,1,1]. Fill the template with chosen symbol-types before any factorial.

C6 | SUM OF ALL NUMBERS WITH REPEATED DIGITS

C6

Example structure: digits 1, 2, 2, 3

1	2	2	3

There are 4!/2! distinct 4-digit numbers. For the total sum, do not list them. Count each digit's total contribution to each place.

H2 POSITION HELPER **Frequency per place**

Fix a place, say the units place.

If 1 is fixed there, the remaining multiset {2,2,3} has 3!/2! arrangements.

If 3 is fixed there, the same count occurs.

If a 2 is fixed, the remaining {1,2,3} has 3! arrangements.

Teacher habit

MODEL **Contribution method**

1. Pick one place value.
2. Count arrangements with each possible digit fixed there.
3. Multiply digit x frequency.
4. Use symmetry of place positions when valid.
5. Multiply by 1111 only after confirming equal total digit contribution per place.

MISCONCEPTION **AVERAGE WITHOUT CHECKING**

Tempting thought: “Each digit appears equally often in each place because permutations are symmetric.”

Why it fails: Repeated copies can still preserve position symmetry, but the frequencies of different digit values need not be equal.

Repair: Count occurrences per digit or use total digit-multiplicity carefully before invoking symmetry.

MISCONCEPTION CLINIC A

Three errors about indistinguishability

MIS A

MISCONCEPTION

USE n! DIRECTLY

Tempting thought: “There are n objects, so n!.”

Why it fails: Identical-copy swaps are invisible, so n! counts the same visible outcome repeatedly.

Repair: Record multiplicities before calculating.

MISCONCEPTION

DIVIDE BY THE WRONG FACTOR

Tempting thought: “Two symbols repeat, so divide by 2!.”

Why it fails: Each repeated group contributes its own invisible internal permutations.

Repair: Divide by a!b!... using each group size.

MISCONCEPTION

TREAT COPIES AS CHOICES

Tempting thought: “Choose which copy of the repeated symbol is used.”

Why it fails: Identical copies are not distinguishable choices unless the problem explicitly labels them.

Repair: Ask whether the recorded outcome can tell the copies apart.

REPAIR HABIT

One-minute diagnostic

Before any formula, write a frequency table:

symbol -> multiplicity

Then ask which swaps are invisible.

MISCONCEPTION CLINIC B

Hybrid repeated-object questions add an inventory stage

MIS B

MISCONCEPTION

FORGET TO CHOOSE THE REPEATED TYPE

Tempting thought: “Pattern 2+1+1 means 4!/2!, done.”

Why it fails: The pattern specifies multiplicities, but not which available symbol occupies the repeated role.

Repair: Choose eligible symbol-types first; arrange second.

MISCONCEPTION

COUNT A DEFERRED RESTRICTION HERE

Tempting thought: “No two repeated symbols together is still just a multiset formula.”

Why it fails: Adjacency introduces gap/block machinery beyond plain repeated-object counting.

Repair: Own the question by its strongest restriction and defer it to the gap/block subtopic.

MISCONCEPTION

DICTIONARY RANK = PLAIN MULTISET

Tempting thought: “Repeated letters mean just divide factorials.”

Why it fails: Dictionary rank requires prefix counting; the multiset formula only counts each prefix block.

Repair: Use repeated-object counting as secondary machinery inside the later lexicographic method.

AUDIT HABIT

Prevent subtopic drift

A question belongs here only when repeated-object counting is the primary engine. If a stronger structure dominates - gaps, blocks, dictionary rank, divisibility - assign primary ownership there.

GUIDED PRACTICE

Scaffolding fades from model to independent setup

GUIDED

A | Complete the multiplicities

TRY

Fill the skeleton

How many distinct arrangements of A, A, A, B, B, C?

Start: total = 6; multiplicities = 3, 2, 1; count = ________.

B | Choose a representation

TRY

Representation control

A 9-slot string has exactly four X, three Y and two Z. Write two equivalent counting expressions: one factorial-division form and one position-selection form.

C | Hybrid inventory

TRY

Do not calculate first

Available symbols include three copies of A, two copies of B, and single copies of C,D,E. Form 4-symbol strings with multiplicity pattern 2+1+1. What must be chosen before arranging?

D | Explain

REASON

Name the assumption

A student says $7!/(2!)$ is always correct whenever exactly one letter repeats. State the missing condition that makes the formula valid.

JEE TRANSFER MAP

Recognise repeated-object counting even when the surface changes

TRANSFER

C3/C4

Prescribed sequence counts

Trigger: 5 ones, 3 twos, 2 zeros

Ask: What multiplicity pattern is fixed, and what remains to be chosen?

C5

Repeated digits selected first

Trigger: one digit repeated twice

Ask: What multiplicity pattern is fixed, and what remains to be chosen?

C5

Word with one repeated pair

Trigger: two alike + two distinct

Ask: What multiplicity pattern is fixed, and what remains to be chosen?

C6

Sum across repeated-digit numbers

Trigger: digits 1,2,2,3

Ask: What multiplicity pattern is fixed, and what remains to be chosen?

METHOD CONTROL

One primary owner per question

Repeated-object counting is often secondary machinery. The audit assigns primary ownership by the strongest structural restriction so the chapter does not duplicate the same PYQ across several subtopics.

MASTERY CHECK + LOCAL COVERAGE AUDIT

Do not advance unless both learning and corpus checks are clean

AUDIT

Can you now...

1. explain why identical swaps are invisible?
2. derive n!/(a!b!...) rather than memorize it?
3. switch between multiset and position-selection views?
4. separate inventory choice from arrangement?
5. handle a repeated-digit positional-sum setup?
6. identify when a stronger restriction owns the question?

Frozen ST02 corpus slice

2025	10-term 0/1/2 sequence	REQUIRED
2026	9-digit repeated-pattern ratio	REQUIRED
2020	SYLLABUS 2+1+1 words	REQUIRED
2023	sum using 1,2,2,3	REQUIRED
2024	BHBJO dictionary rank	DEFER -> Dictionary
2022	colored cubes separated	DEFER -> Gaps

STATUS

Audit stamp

Local ST02 slice: 4 REQUIRED, all mapped into companion practice. Chapter-wide status remains PROVISIONAL until the full 220-question ledger is frozen.

POSITIONAL RESTRICTIONS

Fix the restriction first; count only what is still free

POS 1	POS 2	POS 3	POS 4
R	?	?	E

Essential Question

How do fixed, allowed and relative positions change a permutation count?

THIS SUBTOPIC BUILDS

1. Restriction-first slot marking
2. Allowed-position and category placement
3. Relative-order symmetry

BOUNDARY

No blocks/gaps yet

No divisibility / leading-zero machinery

Repeated objects appear only as secondary tools

HOW THIS SUBTOPIC WORKS

Position restrictions are applied before the free arrangement

ST03

Concept spine

- **C1** — **Restriction-first habit** — mark locked/allowed slots
- **C2** — **Fixed slot / fixed object** — remove what is already decided
- **C3** — **Allowed subset of slots** — choose positions, then arrange
- **C4** — **Only vs exactly** — read logical wording precisely
- **C5** — **Relative order** — symmetry / predetermined order

SCOPE **One primary owner**

If the strongest difficulty is adjacency, use the later block/gap subtopic. If it is divisibility, bounds, or leading zero, use the digit-formation unit.

Pedagogy layers

1. Real-life seat/role constraints
2. Meaning before formula
3. Slot map / allowed-position map
4. Teacher model
5. Guided completion
6. Misconception + repair
7. JEE trigger language
8. Disguised transfer
9. Retrieval + audit

START WITH SEATS THAT ARE NOT INTERCHANGEABLE

Entry A: labelled positions create the restriction

ENTRY A

REAL LIFE

Reserved centre seat

Five selected books are displayed in five labelled shelf positions. One dictionary must occupy the centre. Once the dictionary is placed, only four positions remain free for the novels.

POS 1	POS 2	POS 3	POS 4	POS 5
N	N	D	N	N

REAL LIFE

Badge with locked ends

A four-character event badge must begin with R and end with E. The two internal positions are the only positions still to be filled.

POS 1	POS 2	POS 3	POS 4
R	?	?	E

NOTICE

Restriction before counting

A positional restriction does not add a mysterious formula. It changes which slots are free. Mark the constrained slots first; then count the legal fillings of the remaining slots.

SOME RESTRICTIONS APPLY TO A WHOLE CLASS OF OBJECTS

Entry B: categories can be confined to position sets

ENTRY B

REAL LIFE

Alternating access lanes

Four blue passes may be used only at even-numbered gates 2,4,6,8. If there are exactly four blue passes, the allowed position set is completely filled by blue passes.

1	2	3	4	5	6	7	8
ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD	EVEN

REAL LIFE

Presentation order

Six speakers may appear in any order, but A must speak before E. The restriction is not tied to fixed slot numbers; it is a relative-order condition.

POS 1	POS 2	POS 3	POS 4	POS 5	POS 6
?	A	?	?	E	?

SYNTHESIS

Three restriction shapes

Positional restrictions come in different shapes: fixed slot, allowed slot set, and relative order. The first job is to classify the shape before counting.

C1 | MARK THE RESTRICTION BEFORE ANY FACTORIAL

C1

Teacher habit

MODEL

Restriction-first algorithm

1. Draw labelled positions.
2. Lock any fixed positions.
3. Shade allowed position sets.
4. Record objects already consumed.
5. Only then count the remaining legal fillings.

POS 1	POS 2	POS 3	POS 4
R	_	_	E

WHY IT WORKS

Model before notation

A formula such as nPr describes a free ordered selection. Once positions have special rules, the slot picture prevents you from accidentally counting illegal assignments.

JEE trigger language

1. first / last / middle
2. must occupy position k
3. only in odd/even places
4. A before B / alphabetical order
5. fixed at the ends
6. a category restricted to selected places

CHECK

One diagnostic question

Before solving, say aloud: “Which positions are restricted, and which objects are consumed by that restriction?”

C2

Example: one dictionary in the middle

POS 1	POS 2	POS 3	POS 4	POS 5
N	N	D	N	N

After the dictionary is chosen and placed in the centre, the four remaining shelf positions are filled only by the selected novels.

choose dictionary x choose novels x arrange 4 free slots

CORE

Remove decided structure

A fixed object is not included again in the factorial for the free positions. Fix it once; then count only what remains.

Misconception repair

MISCONCEPTION

FIXED ITEM COUNTED TWICE

Tempting thought: “Choose the dictionary, fix it in the middle, then arrange all five chosen books in 5! ways.”

Why it fails: The 5! step moves the dictionary away from the centre in most arrangements, undoing the restriction you already enforced.

Repair: After fixing an object, remove both that object and that position from the free arrangement.

H2 VISUAL

Cross-out test

Cross out the locked position and the object occupying it. The remaining count should involve only uncrossed objects and uncrossed positions.

C3 | A CATEGORY MAY USE ONLY A SUBSET OF POSITIONS

Choose legal positions first when the subset is not fully forced

C3

Model: 3 odd digits, 4 even-numbered positions

1	2	3	4	5	6	7	8	9
ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD

There are four even-numbered positions but only three odd digits. “Odd digits occupy even places” means choose which three of those four positions receive the odd digits.

choose 3 of 4 allowed slots -> arrange odd multiset -> fill the rest

CORE

Choose positions, then arrange

When objects are restricted to an allowed position set that has more slots than objects, a position-choice stage appears before arrangement.

Contrast: 4 even digits, 4 even positions

1	2	3	4	5	6	7	8	9
ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD

If exactly four even digits must occupy only the four even-numbered positions, there is no choice of slot subset: those four positions are forced.

COMPARE

Same wording, different count shape

3 restricted objects into 4 allowed slots:
choose a 3-slot subset.

4 restricted objects into 4 allowed slots:
all allowed slots are forced.

This distinction drives the 2019 vs 2023 PYQs.

C4 | READ “ONLY” AND “EXACTLY” PRECISELY

Logic words decide whether unused allowed positions remain

C4

Only = forbidden outside the set

MEANING

Allowed-set statement

“Even digits occupy only even places” means an even digit may not appear in an odd-numbered position. It does not, by wording alone, say every even position must contain an even digit.

CHECK

Count comparison

To know whether every allowed slot is actually used by the restricted category, compare:

number of restricted objects vs number of allowed slots.

Exactly = quantity is fixed

MEANING

Quantity statement

“Exactly three odd digits are in even positions” fixes the number of restricted objects using that position set. A subset-choice stage is normally required if more than three even positions exist.

MISCONCEPTION

ONLY = EXACTLY

Tempting thought: “Only even places” means every even place is filled by that category.

Why it fails: Only restricts where the category may occur; whether all allowed slots are occupied depends on how many such objects exist.

Repair: Count the category and the allowed slots before deciding whether a slot-selection factor is needed.

C5 | RELATIVE ORDER IS NOT A FIXED SLOT

Use symmetry when only internal order is prescribed

C5

Example: GARDEN with vowels alphabetical

The letters are all distinct. The vowels are A and E. In every arrangement where A appears before E, swapping only A and E produces a partner arrangement where E appears before A.

POS 1	POS 2	POS 3	POS 4	POS 5	POS 6
G	A	R	D	E	N

POS 1	POS 2	POS 3	POS 4	POS 5	POS 6
G	E	R	D	A	N

exactly half of all 6! arrangements have A before E

CORE

Relative-order symmetry

If k distinct selected objects must appear in one prescribed relative order, symmetry often contributes a factor of 1/k!, provided no other condition breaks that symmetry.

Misconception repair

MISCONCEPTION

FIXED-POSITION THINKING

Tempting thought: “A before E means choose a fixed position for A, then a later position for E.”

Why it fails: That can work, but it creates an unnecessary case sum. The restriction concerns only their relative order, not exact positions.

Repair: Look for a swap symmetry: among the 2! internal A/E orders, exactly one is allowed.

JEE TRIGGER

Language cue

“alphabetical order”, “in increasing order”, “A precedes B”, and “relative order unchanged” should trigger an internal-order check.

C6 | REPEATED OBJECTS + POSITION RESTRICTIONS

Lock the positions first, then apply ST02 inside each free group

C6

2023-style pattern

1	2	3	4	5	6	7	8	
ODD	EVEN	ODD	EVEN	ODD	EVEN	ODD	EVEN	OI

Digits of 123412341 have multiplicities: 1x3, 2x2, 3x2, 4x2. The four even digits (2,2,4,4) must use only the four even-numbered positions.

even slots: 4!/(2!2!) | odd slots: 5!/(3!2!)

ORDER OF OPERATIONS

Restriction first, repeats second

1. Identify restricted category.
2. Identify its allowed slots.
3. Decide whether the slot set is forced.
4. Arrange the restricted multiset there.
5. Arrange the remaining multiset in the remaining slots.

Why the order matters

MISCONCEPTION

DIVIDE FIRST, RESTRICT LATER

Tempting thought: “Start with 9!/(3!2!2!2!) and then somehow remove numbers with misplaced even digits.”

Why it fails: That complement is possible but structurally messy. The restriction cleanly partitions the positions into even and odd sets.

Repair: Exploit the position partition first; then use repeated-object formulas within each group.

TRANSFER

Partition-and-arrange

Whenever a position restriction splits the objects into groups, the final count often becomes a product of independent multiset arrangements inside those groups.

TWO JEE MODELS, SAME RESTRICTION-FIRST HABIT

One fixed-end word and one fixed-middle shelf

MODELS

Model A | first R, fourth E

POS 1	POS 2	POS 3	POS 4
R	?	?	E

Use one R and one E first. The middle two slots are then filled from the remaining letter inventory. Count different ordered letter-types plus allowed identical pairs; do not treat repeated copies as distinct.

THINK **JEE Main 2016**

Remaining distinct types: M,E,D,I,T,R,A,N.

Different middle letters: 8x7.
Same-letter pair allowed only where at least two copies remain: E, A, N.

Total skeleton: 8x7 + 3.

OPEN 2016 PYQ

Model B | dictionary fixed in middle

POS 1	POS 2	POS 3	POS 4	POS 5
N	N	D	N	N

There is an inventory stage before arrangement: select 4 novels from 6 and one dictionary from 3. The dictionary is then locked in the middle; only the 4 novels are permuted.

THINK **JEE Main 2018**

Choose 4 novels: C(6,4).
Choose dictionary: 3.
Fix dictionary in centre: 1 way.
Arrange novels in remaining positions: 4!.

Multiply sequential stages.

OPEN 2018 PYQ

MISCONCEPTION CLINIC A

Three errors about fixed and allowed positions

MIS A

MISCONCEPTION

FORMULA FIRST

Tempting thought: “It says arrangement, so write n! or nPr immediately.”

Why it fails: The restriction changes which assignments are legal before the free count begins.

Repair: Draw slots and mark restrictions before choosing a compact formula.

MISCONCEPTION

FIXED ITEM DOUBLE COUNT

Tempting thought: “After fixing an item, still permute all chosen objects.”

Why it fails: That moves the item away from its forced position in most counted arrangements.

Repair: Remove the fixed item and fixed slot from the free problem.

MISCONCEPTION

POSITION SUBSET MISSED

Tempting thought: “Three odd digits go to even places, so just arrange them in 3! ways.”

Why it fails: There may be more allowed even positions than restricted digits; the positions themselves must first be chosen.

Repair: Compare restricted-object count with allowed-slot count.

REPAIR HABIT

Three-column diagnostic

Write under the slot map:

LOCKED? | ALLOWED SET? | HOW MANY OBJECTS?

The correct count shape usually becomes obvious.

MISCONCEPTION CLINIC B

MIS B

MISCONCEPTION

ONLY = EXACTLY

Tempting thought: “Only even positions” means all even positions must be filled by even digits.

Why it fails: Only forbids placement outside the allowed set; complete filling depends on the inventory count.

Repair: Count objects and allowed slots before deciding whether the set is forced.

MISCONCEPTION

REPEATS FORGOTTEN AFTER RESTRICTION

Tempting thought: “Once parity positions are fixed, arrange each group with ordinary factorials.”

Why it fails: Repeated digits remain indistinguishable inside their position group.

Repair: Apply ST02 multiset division within each restricted group.

MISCONCEPTION

RELATIVE ORDER OVERCASED

Tempting thought: “A before E requires many cases for exact positions.”

Why it fails: The restriction may only select one of the 2! internal A/E orders, so symmetry is much shorter.

Repair: Test whether swapping the restricted objects pairs every legal and illegal order.

AUDIT HABIT

Prevent drift

If a question also contains adjacency, divisibility, leading-zero, dictionary rank, or circular equivalence, assign primary ownership to the stronger structure and keep ST03 as secondary machinery.

GUIDED PRACTICE

Scaffolding fades from fixed slots to hybrid restrictions

GUIDED

A | Fixed slot

TRY

Fill the skeleton

Seven distinct people stand in a row. A must be in position 4. Complete the count:

A fixed -> ______ free people in ______ free slots -> ______ arrangements.

B | Allowed slot set

TRY

Name the missing stage

Three red cards must occupy odd-numbered positions among 7 labelled positions. There are four odd positions. What stage must occur before arranging the red cards?

C | Only vs exactly

REASON

Inventory vs allowed slots

A 9-slot sequence contains exactly four X symbols. X may appear only in even positions. Are the X positions forced? Explain without calculating.

D | Relative order

REASON

Internal-order symmetry

Eight distinct runners are listed in random order. A, B and C must appear in alphabetical relative order. What symmetry factor should you expect, and why?

JEE TRANSFER MAP

Recognise the restriction shape before the surface wording

TRANSFER

C2/C6

Fixed ends of a word

Recognition: first R, fourth E

Ask: Which slots are fixed, selectable, or merely ordered?

C2

Middle object on shelf

Recognition: dictionary always middle

Ask: Which slots are fixed, selectable, or merely ordered?

C3/C4/C6

Odd digits in even places

Recognition: 3 objects -> 4 allowed slots

Ask: Which slots are fixed, selectable, or merely ordered?

C3/C4/C6

Even digits only even places

Recognition: 4 objects -> 4 allowed slots

Ask: Which slots are fixed, selectable, or merely ordered?

C5

Vowels alphabetical

Recognition: relative order only. Ask: Is this a relative-order symmetry rather than fixed slots?

MASTERY CHECK + LOCAL COVERAGE AUDIT

Advance only when learning and source coverage both reconcile

AUDIT

Can you now...

1. mark fixed and allowed slots before counting?
2. remove a fixed item from the free factorial?
3. detect when an allowed-slot subset must be chosen?
4. distinguish “only” from “exactly”?
5. use relative-order symmetry when appropriate?
6. combine ST02 repeats after position locking?

Frozen ST03 corpus slice

2016	MEDITERRANEAN fixed R/E	REQUIRED
2018	dictionary fixed middle	REQUIRED
2019	odd digits -> even places	REQUIRED
2023	even digits only even places	REQUIRED
2004	GARDEN vowels alphabetical	REQUIRED
2009	same dictionary-middle fingerprint	EXCLUDE DUPLICATE

STATUS

Audit stamp

Local ST03 slice: 5 REQUIRED, all mapped to companion practice. Chapter-wide status remains PROVISIONAL until the complete 220-row ledger is classified.

TOGETHER: THE BLOCK METHOD

Compress a consecutive group into one outer object - then restore its internal order

GLUE THESE TOGETHER

A B C -> [ABC]

Essential Question

When several objects must stay consecutive, what exactly changes in the arrangement?

THIS SUBTOPIC BUILDS

1. Glue -> arrange blocks -> unglue
2. Internal permutations inside a block
3. Repeated-object blocks and multiple blocks

BOUNDARY

ST05 owns “not together” / complement

ST06 owns “no two together” / gaps

ST04 owns positive consecutiveness

HOW THIS SUBTOPIC WORKS

One block is one outer object, but its inside may still move

ST04

Concept spine

C1 **Recognise a block** — together / consecutive

C2 **Outer vs inner order** — arrange blocks x arrange inside

C3 **Repeated objects** — multiset outside and inside

C4 **Multiple blocks** — each glued group is one object

C5 **Block + position rule** — place the block as a unit

SCOPE **Do not mix engines**

A block is created only by a positive consecutiveness condition. “Not together” is counted later by complement; “no two together” is a gap problem.

Pedagogy layers

1. **Real-life consecutive grouping**
2. **Glue visual before formula**
3. **Outer/inner two-layer model**
4. **Teacher example**
5. **Misconception + repair**
6. **JEE trigger wording**
7. **Repeated-object transfer**
8. **Mixed-owner classification**
9. **Retrieval + audit**

START WITH GROUPS THAT MUST STAY PHYSICALLY ADJACENT

Entry A: consecutive seats create a super-object

ENTRY A

REAL LIFE

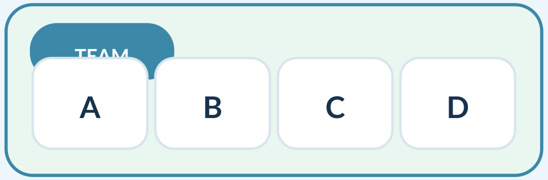
Family photo

Seven people line up for a photo. Three siblings must stand next to one another as a group. From the outside, the three siblings move like one large object.

REAL LIFE

Reserved train seats

A four-person team must occupy four consecutive seats in a row. The team's internal seat order can change even though the team remains one consecutive block.



NOTICE

Two-layer structure

The restriction has two levels: outside, the group behaves as one object; inside, its members may still permute. A correct count must respect both levels.

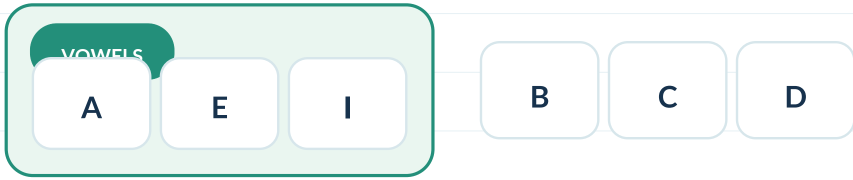
THE SAME IDEA APPEARS IN WORDS

Entry B: vowels or selected letters can become one block

ENTRY B

Word example

Suppose all vowels in a word must occur together. Do not place the vowels one by one into arbitrary positions. Glue the entire vowel group first.



ABSTRACT ONLY NOW

Glue -> outer -> inner

If the vowel block plus the remaining consonants produce m outer objects, arrange those outer objects first. Then multiply by the number of distinct arrangements inside the vowel block.

What "together" means

CONCEPT

One maximal block

"All A-type objects together" means they occupy one consecutive run. It does not mean that every pair is separately treated as a block.

MISCONCEPTION

PAIRWISE GLUING

Tempting thought: "If three vowels are together, make blocks AE, EI and AI."

Why it fails: Those blocks overlap and do not represent a legal decomposition of one arrangement.

Repair: Create exactly one block containing all objects required to be consecutive

JEE TRIGGER

Language cue

Words such as all together, always together, consecutive, in one group, adjacent as a group suggest a block.

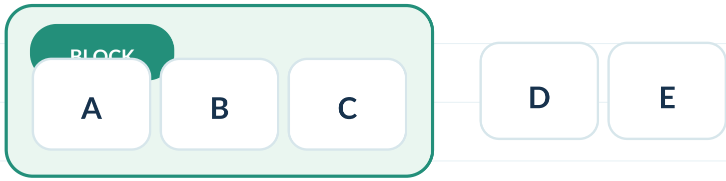
C1 | GLUE FIRST: COUNT OUTER OBJECTS

A group of k consecutive objects reduces the outer count by k-1

C1

Teacher model

Arrange A, B, C, D, E in a row so that A, B and C stay together.



outer objects: [ABC], D, E -> 3!

FIRST COUNT

Compression

At the outer level there are not 5 objects anymore. The three glued objects have become one super-object, so the outer arrangement uses 3 objects.

General compression rule

MODEL

Count objects, not symbols

Start with n distinct objects. If a specified group of k objects must be consecutive, replace those k objects by one block.

Outer object count = n - k + 1.

n objects -> n-k+1 outer objects

MISCONCEPTION

REMOVE k OBJECTS

Tempting thought: “The block removes k objects, so outer count is n-k.”

Why it fails: The glued group does not disappear; it becomes one new outer object.

Repair: Subtract k, then add one block back: n-k+1.

Outer arrangement and internal arrangement are independent stages

C2

Continue the same example

[ABC]	D	E

OUTER **Stage 1**

Arrange the three outer objects [ABC], D, E in 3! ways. At this stage ABC is still one object.

INNER **Stage 2**

Inside the block, A, B and C can appear in 3! orders. These internal orders can occur for every legal outer arrangement, so multiply.

The full block skeleton

outer arrangements x internal arrangements

3! x 3!

WHY MULTIPLY **Product rule returns**

Choose an outer position of the block and then choose one internal order. Every outer arrangement supports every internal arrangement.

MISCONCEPTION **FORGET INNER ORDER**

Tempting thought: “Once ABC is a block, the answer is just 3!.”

Why it fails: That treats ABC as a rigid printed symbol and loses the 3! possible orders of A, B and C inside it.

Repair: After every block count, explicitly ask: can the inside rearrange?

C3

Example structure: INDEPENDENCE

Vowels are I,E,E,E,E. Consonants are N,N,N,D,D,P,C. If all vowels stay together, build one vowel block V.

V

| I | E | E | E | E |

OUTER MULTISET

ST02 outside the block

Outer objects are V, N,N,N, D,D, P, C: 8 objects, with N repeated 3 times and D repeated twice.

Outer count = 8!/(3!2!).

INNER MULTISET

ST02 inside the block

Inside V, the five vowels I,E,E,E,E have E repeated four times.
Internal count = 5!/4!.

Where students go wrong

MISCONCEPTION

DIVIDE ONLY ONCE

Tempting thought: “Repeated letters are already handled outside, so no division is needed inside the vowel block.”

Why it fails: Repeated E’s live inside the block; repeated N’s and D’s live among outer objects. These are separate symmetry corrections.

Repair: Write one inventory for the outer level and another for the block interior.

CONCEPT

Layer-by-layer counting

Block formation does not erase repeated-object structure. Apply the correct multiset count separately at each independent level.

JEE TRIGGER

Hybrid cue

Together + repeated letters is a signal to combine ST04 block with ST02 multiset.

C4 | MULTIPLE BLOCKS: EACH REQUIRED GROUP BECOMES ONE OBJECT

Two positive together-conditions create two super-objects

C4

Real-life model

Three girls must stay together and four boys must stay together in one queue. Ignore any extra restrictions for the moment.

GIRLS BLOCK

G1 G2 G3

BOYS BLOCK

B1 B2 B3 B4

outer: 2! | inner: 3! x 4!

STRUCTURE

One factor per level

At the outer level there are only two objects: G-block and B-block. The internal arrangements of each block are independent, so the unrestricted two-block count is 2! x 3! x 4!.

General rule

MODEL

Multiple blocks

For several disjoint required groups:

1. glue every required group,
2. count the resulting outer objects,
3. multiply by each block's internal arrangements.

MISCONCEPTION

ONE GIANT BLOCK

Tempting thought: "If two groups must each stay together, glue all of them into one block."

Why it fails: That incorrectly forces the two groups to touch each other and destroys legal arrangements where other free objects lie between them.

CHECK

Outer order still matters

If two blocks are present with no free objects, can their order swap? If yes, the outer factor includes 2!.

C5 | A BLOCK CAN ALSO HAVE A POSITION RESTRICTION

Treat the block itself as one object with allowed outer positions

C5

Example

Eight distinct people stand in a row. A, B and C must be consecutive, and

H1 **Translate condition**

First form the block [ABC]. Then treat the statement about where it may start as a position restriction on the block, not on A alone.

H2 **Block-as-object view**

Possible block spans are positions 1-3, 2-4, or 3-5. For each allowed span, arrange A,B,C internally and arrange the five remaining people in the five remaining positions.

BOUNDARY **Primary/secondary owner**

This is ST04 + ST03. The positive consecutive group still owns the problem; ST03 only restricts where the resulting super-object may be placed.

Concept control

MISCONCEPTION **POSITION A ONLY**

Tempting thought: “If the block starts early, just restrict A to the first three positions.”

Why it fails: A need not be the first internal member of the block; any of A,B,C may appear first within the consecutive run.

Repair: Restrict the block start, then restore all internal orders.

JEE TRIGGER **Hybrid cue**

Phrases such as together and at an end, consecutive starting from..., or

REPAIR **Visual discipline**

Whenever a block has an extra condition, draw the large block rectangle first. Attach the condition to that rectangle before arranging its contents.

C6

Three similar-looking prompts

ST04 BLOCK

All 3 girls together

glue girls

ST05 COMPLEMENT

All 3 girls not together

total - all together

ST06 GAPS

No two girls together

arrange others, use gaps

READ WORDING

Language precision

“Not all together” and “no two together” are not synonyms. They produce different forbidden sets and therefore different counting engines.

Mixed 2025-style queue

Girls form one block and boys form one block, but B1 and B2 are not adjacent inside the boys block.

PRIMARY

Two outer blocks

The outer architecture is created by two positive together conditions, so ST04 is the primary owner.

SECONDARY

Nested method

Inside the boys block, the B1/B2 restriction is a local ST05 complement count: all boy orders minus adjacent-pair boy orders.

AUDIT RULE

Prevent drift

A mixed PYQ may be included here if the block architecture is indispensable and dominant. Otherwise it is deferred to the subtopic that owns the strongest constraint.

TWO JEE MODELS: BLOCK + REPEATED OBJECTS

Same engine, different inventories

MODELS

Model A | INDEPENDENCE

Vowels: I,E,E,E,E. Consonants: N,N,N,D,D,P,C.

OUTER

Block among repeated consonants

Glue the five vowels into V. Arrange V,N,N,N,D,D,P,C:

8!/(3!2!)

INNER

Repeated E inside block

Arrange I,E,E,E,E inside V:

5!/4!

[8!/(3!2!)] x [5!/4!] = 16800

OPEN 2023 PYQ

Model B | ASSASSINATION

Vowels: A,A,A,I,I,O. Consonants: S,S,S,S,N,N,T.

OUTER

Outer multiset

Glue the six vowels into V. Arrange V,S,S,S,S,N,N,T:

8!/(4!2!)

INNER

Inner multiset

Arrange A,A,A,I,I,O inside V:

6!/(3!2!)

[8!/(4!2!)] x [6!/(3!2!)] = 50400

OPEN 2023 PYQ

MISCONCEPTION CLINIC A

Three failures in basic block construction

MIS A

MISCONCEPTION

FORGET INTERNAL ORDER

Tempting thought: “Glue ABC, then arrange only the outer objects.”

Why it fails: The members inside the block can usually rearrange while remaining consecutive.

Repair: Multiply by the distinct internal arrangements of the block.

MISCONCEPTION

OUTER COUNT OFF BY ONE

Tempting thought: “Replacing k objects removes k positions, so use n-k outer objects.”

Why it fails: The block itself is still an object in the outer arrangement.

Repair: Outer object count is n-k+1 for one block of k specified objects.

MISCONCEPTION

OVERLAPPING PAIR BLOCKS

Tempting thought: “Three vowels together means create several adjacent pairs.”

Why it fails: Pair blocks overlap and double-count; together means one consecutive run.

Repair: Use one block containing every object required to stay together.

REPAIR HABIT

Two inventories

Write two lines before calculation:

OUTER INVENTORY: ______
INNER INVENTORY: ______

MISCONCEPTION CLINIC B

Repeated objects and mixed restrictions

MIS B

MISCONCEPTION

REPEATS DIVIDED AT WRONG LEVEL

Tempting thought: “Divide by every repeated factorial once in the final expression.”

Why it fails: Some repeats are inside a block and others are among outer objects; the symmetries belong to different stages.

Repair: Apply multiset division separately to each inventory.

MISCONCEPTION

TWO BLOCKS MERGED

Tempting thought: “Girls together and boys together means all seven people form one rigid block.”

Why it fails: The two groups are independently consecutive and may swap order; they are not one frozen object.

Repair: Create one block per required group, then arrange the blocks.

MISCONCEPTION

NOT TOGETHER SOLVED AS BLOCK

Tempting thought: “The phrase contains ‘together’, so use block directly.”

Why it fails: A negative condition asks for the complement of the together event; the block only counts the forbidden/removed part.

Repair: Move the whole problem to ST05 unless a positive block is the dominant architecture.

AUDIT HABIT

Owner test

Tag the wording first:

positive together -> ST04
not all together -> ST05
no two together -> ST06

GUIDED PRACTICE

Scaffolding fades from one block to repeated-object blocks

GUIDED

A | One distinct block

TRY

Fill both levels

Six distinct people stand in a row. A, B and C must be together.

Outer objects = ______
Outer count = ______
Inner count = ______

B | Two blocks

TRY

One block per group

Three teachers must stay together and four students must stay together. No other restriction. Write the product skeleton without evaluating.

C | Repeated letters

REASON

Inventories first

A word has vowels A,A,E and consonants N,N,R,T. All vowels must be together. Write the outer and inner inventories before any factorial.

D | Owner classification

REASON

Method control

Classify each: (i) all girls together, (ii) all girls not together, (iii) no two girls together. Name ST04/ST05/ST06 and the first structural move.

JEE TRANSFER MAP

Recognise the block fingerprint under different surface stories

TRANSFER

C3

Repeated-letter word

Recognition: all vowels together

First move: draw one large block around the required consecutive group.

C1/C2

People in queue

Recognition: one family stays together

First move: draw one large block around the required consecutive group.

C4

Two groups

Recognition: girls together + boys together

First move: draw one large block around the required consecutive group.

C5

Block with fixed span

Recognition: consecutive + position rule

First move: draw one large block around the required consecutive group.

C6

Mixed internal restriction

Recognition: blocks outside; complement inside. Control: keep ST04 as primary owner only when the outer block architecture is indispensable.

MASTERY CHECK + LOCAL COVERAGE AUDIT

Advance only when the block model and source slice both reconcile

AUDIT

Can you now...

1. recognise positive consecutiveness as a block?
2. count n-k+1 outer objects for one specified k-block?
3. restore internal order after the outer arrangement?
4. apply repeated-object division at the correct level?
5. create several independent blocks correctly?
6. separate ST04 from complement and gap owners?

Frozen ST04 corpus slice

2023	INDEPENDENCE vowels together	REQUIRED
2023	ASSASSINATION vowels together	REQUIRED
2025	girls block + boys block + local nonadjacency	REQUIRED HYBRID
2025	DAUGHTER vowels never together	DEFER -> ST05
2026	all girls not together	DEFER -> ST05
2025	all boys OR no two boys	DEFER -> mixed ST04/ST06

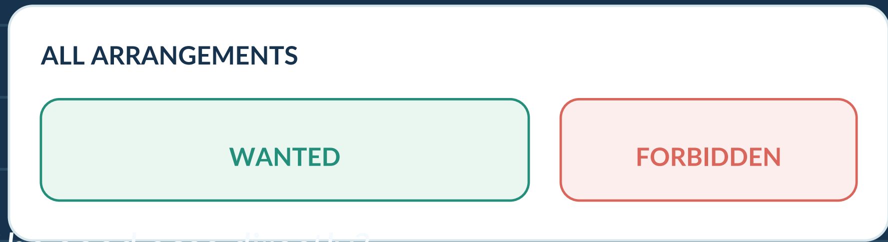
STATUS

Audit stamp

Local ST04 slice: 3 REQUIRED, all mapped to companion practice. Chapter-wide status remains PROVISIONAL until the complete 220-row ledger is classified.

NOT TOGETHER: COMPLEMENT

Count the whole universe, count the forbidden together-event, subtract once



Essential Question

When is "count everything except the bad case" safer than counting the good case directly?

THIS SUBTOPIC BUILDS

1. Total - forbidden event
2. Pair nonadjacency and "not all together"
3. Repeated-object complements and nested use

BOUNDARY

ST04 owns positive together

ST06 owns no-two-together / gaps

Later IE handles overlapping forbidden events

HOW THIS SUBTOPIC WORKS

One clean forbidden event -> complement is usually the shortest route

ST05

Concept spine

C1	**Complement partition**	wanted = total - forbidden
C2	**Two specified objects apart**	forbidden = one adjacent block
C3	**A whole group not all together**	subtract one all-together event
C4	**Repeated objects**	multiset total - multiset bad
C5	**Nested complement**	apply inside a larger architecture

SCOPE

Boundary

ST05 assumes the complement contains a manageable forbidden event. If several forbidden events overlap, simple subtraction may be incomplete.

Pedagogy layers

1. **Real-life exclusion story**
2. **Universe partition visual**
3. **Block used only for BAD event**
4. **Teacher model**
5. **Misconception + repair**
6. **JEE trigger wording**
7. **Repeated-object transfer**
8. **Nested restriction**
9. **Retrieval + audit**

START WITH “EVERYTHING EXCEPT THIS”

Entry: the complement is a partition of the same universe

ENTRY

REAL LIFE

Presentation order

Five speakers present once each. Two rival speakers A and B must not present consecutively. Instead of constructing every legal schedule, count all schedules and remove those where A and B form one adjacent block.

A B -> [AB] C D E

REAL LIFE

Three VIPs in a queue

Seven people stand in one queue. Three specified VIPs are not allowed to occupy one single consecutive run. The forbidden event - all three togeth

ALL ARRANGEMENTS

WANTED FORBIDDEN

NOTICE

Partition the universe

Complement works because wanted and forbidden split the same complete universe with no overlap: every arrangement is either valid or forbidden.

C1 | THE COMPLEMENT ENGINE

Define TOTAL and BAD before subtracting

C1

The three-line discipline

1. UNIVERSE TOTAL

Write the full set of arrangements that obey all background rules but ignore the one negative condition.

2. FORBIDDEN BAD

Translate the negative condition into a positive event you can count: adjacent, all together, specified pair together, etc.

3. SUBTRACT GOOD

Wanted = TOTAL - BAD. Subtract exactly from the same universe you defined in Step 1.

Concept helper

ALL ARRANGEMENTS

WANTED | FORBIDDEN

GOOD = TOTAL - BAD

MISCONCEPTION WRONG UNIVERSE

Tempting thought: “Use n! as TOTAL even though the problem already has another restriction.”

Why it fails: Then BAD and TOTAL are counted under different background rules, so subtraction no longer partitions one universe.

Repair: Carry every background restriction into both TOTAL and BAD.

C2 | TWO SPECIFIED OBJECTS MUST NOT BE ADJACENT

Count the adjacent pair as the forbidden block

C2

Teacher model: A and B among six people

Six distinct people stand in a row. A and B must not be adjacent.

TOTAL **Same universe**

Ignore the negative condition: arrange all six distinct people.

TOTAL = 6!

BAD **Forbidden adjacency**

Force A and B together. Treat [AB] as one outer object with the other four people: 5! outer orders. Inside, AB can be AB or BA: 2 orders.

GOOD = 6! - 2 x 5!

Visual helper

A B -> [AB] C D E F

WHY x2? **Internal block order**

The block [AB] is not rigid. The forbidden event includes both AB and BA inside the same adjacent pair. Missing this factor undercounts BAD.

JEE TRIGGER **Language cue**

Specified letters/people do not come together, are not adjacent, must be separated. For one forbidden pair, complement is often immediate.

C3 | “NOT ALL TOGETHER” IS NOT “NO TWO TOGETHER”

These phrases describe different forbidden sets

C3

Not all three together

For A, B, C, the condition only forbids the single event that all three occupy one consecutive block. Two of them may still be adjacent.

ALLOWED

Complement owner: ST05

A B X C ... is allowed.
A X B C ... is allowed.

Only configurations containing one consecutive [ABC]-type run are removed.

No two of the three together

Now every adjacent pair among A, B, C is forbidden. The arrangement must actively separate the special objects.

FORBIDDEN

Different event

A B X C ... is now invalid.
A X B C ... is also invalid.

There are several overlapping adjacency events, so one simple block subtraction is not enough.

COUNTING MOVE

Reuse ST04 inside complement

TOTAL - (all three together). The BAD event is a single positive block event from ST04.

OWNER

Do not blur the phrases

Use the gap method in ST06 when a whole group must be mutually separated.

C4

Teacher model: MATHEMATICS, C and S apart

MATHEMATICS has 11 letters with M, A and T each repeated twice. C and S are distinct singletons.

TOTAL

Multiset universe

All distinct arrangements:

11! / (2! 2! 2!)

BAD

Adjacent event

Glue C and S. The block can be CS or SC. Outer inventory has 10 objects with M, A, T still repeated twice:

2 x 10!/(2!2!2!)

GOOD = 11!/(2!2!2!) - 2 x 10!/(2!2!2!)

Concept helper: two inventories

TOTAL INVENTORY **11 symbols**

M,M,A,A,T,T,H,E,I,C,S

BAD INVENTORY **10 outer objects**

M,M,A,A,T,T,H,E,I,[CS]
with [CS] internally reversible

MISCONCEPTION

DIVIDE REPEATS ONLY ONCE

Tempting thought: “Find total, then subtract 2 x 10! without repeat-division.”

Why it fails: The BAD universe still contains repeated M, A and T symbols. Their indistinguishable swaps remain symmetries.

Repair: Rebuild the inventory for BAD instead of modifying TOTAL mentally.

C5 | A WHOLE GROUP IS NOT ALL TOGETHER

One forbidden group-block can still be easy

C5

Model: 4 boys and 3 girls

Seven distinct students form a queue. The condition says all the girls are not together. This means the three girls must not occupy one single

TOTAL

Ignore only the negative condition

All seven distinct students:

7!

BAD

All girls together

Treat the three girls as one block. With four boys, there are five outer objects: 5! outer orders. Inside the girls block: 3! orders.

GOOD = 7! - 5! x 3!

Wording diagnosis

READ CAREFULLY

Logical scope

“All the girls are not together” in this ExamSIDE problem is handled as not(all girls together), not “no two girls adjacent.”

PERMITTED

Allowed configurations

Two girls may still stand adjacent, provided the third girl is elsewhere. That is why a gap-method count would solve a stricter problem than asked.

MISCONCEPTION

NOT ALL = NO TWO

Tempting thought: “If all girls are not together, separate every pair of girls.”

Why it fails: That removes arrangements with two adjacent girls even though the question only forbids all three forming one block.

Repair: Translate the sentence into an event before choosing the method.

C6 | COMPLEMENT CAN LIVE INSIDE A LARGER STRUCTURE

Keep background architecture fixed, subtract locally

C6

Example: two outer blocks

Girls must form one block and boys one block, but two specified boys B1 and B2 must not be adjacent inside the boys block.

OUTER ARCHITECTURE

Background universe

ST04 creates two outer objects: G-block and B-block. Their order and the girls' internal order are counted normally.

LOCAL COMPLEMENT

Nested engine

Inside the 4-boy block:

valid boy orders = 4! - 2 x 3!

Only this internal factor uses ST05.

Expert control

RULE

Local universe

Complement subtracts within the current universe. If another restriction has already created a smaller universe, keep it fixed while counting TOTAL and BAD locally.

OWNER

Audit discipline

Primary owner remains the method that creates the dominant architecture. In this mixed queue, ST04 owns the two blocks; ST05 is a nested substep.

TRANSFER

Same engine, new surface

The same pattern appears in number formation: first restrict the leading digit or parity, then subtract a forbidden adjacency/pattern inside that already restricted universe.

WHEN SIMPLE COMPLEMENT STOPS BEING SIMPLE

One bad event is ideal; several overlapping bad events need more machinery

BOUNDARY

Complement is excellent when...

1. one specified pair must not be adjacent
2. one specified group must not be all together
3. one forbidden block/event has a clean count
4. TOTAL is easy under all background rules

Do not stop at one subtraction when...

1. no two among many special objects may be adjacent
2. several different forbidden pairs can overlap
3. circular symmetry is part of the universe
4. dictionary-order filtering is the dominant task

HEURISTIC

Method selection

If BAD is easier to describe positively than GOOD is to construct directly, complement is a strong candidate.

DEFER

Prevent drift

Gap method, circular permutations, lexicographic counting and inclusion-exclusion receive their own subtopics.

THREE JEE MODELS: SAME COMPLENT ENGINE

Queue, distinct word, repeated-letter word

MODELS

2026 queue

MODEL

Group complement

4 boys + 3 girls; all girls are not together.

7! - 5! x 3! = 4320

OPEN 2026 PYQ

2025 DAUGHTER

MODEL

Distinct word

8 distinct letters; vowels A,U,E not all together.

8! - 6! x 3! = 36000

OPEN 2025 PYQ

2023 MATHEMATICS

MODEL

Repeated-object pair complement

C and S must not be adjacent. Repeats: M2,A2,T2.

TOTAL = 11!/(2!2!2!)
BAD = 2 x 10!/(2!2!2!)
GOOD = 4,082,400 = 6! x 5670

OPEN 2023 PYQ

COMMON DNA

Total - bad

All three solutions begin by writing the same universe, then turn the negative wording into one countable positive event.

MISCONCEPTION CLINIC A

Three failures in defining TOTAL and BAD

MIS A

MISCONCEPTION

NOT ALL = NO TWO

Tempting thought: “Not all girls together means every pair of girls must be separated.”

Why it fails: The phrase forbids one all-together event, not every local adjacency.

Repair: Write the forbidden event explicitly in parentheses: NOT(all three together).

MISCONCEPTION

WRONG TOTAL

Tempting thought: “Use n! as total even after another restriction has already been imposed.”

Why it fails: TOTAL and BAD must be subsets of the same background universe.

Repair: Freeze the background rules before counting either term.

MISCONCEPTION

FORGOT INTERNAL BLOCK ORDER

Tempting thought: “BAD for A,B adjacent is (n-1)!.”

Why it fails: The forbidden block can usually be AB or BA.

Repair: Multiply by the internal block arrangements.

REPAIR HABIT

Three-line discipline

Before arithmetic, write:

Universe: ______
Forbidden event: ______
GOOD = TOTAL - BAD

MISCONCEPTION CLINIC B

Repeated objects and overlapping events

MIS B

MISCONCEPTION

REPEATS LOST IN BAD

Tempting thought: “After forming the forbidden block, every remaining object is distinct.”

Why it fails: Repeated letters elsewhere remain indistinguishable in the BAD inventory.

Repair: Re-list the BAD inventory and divide its own repeated factorials.

MISCONCEPTION

SUBTRACTED TOO MUCH

Tempting thought: “If two girls are adjacent, that must be bad for ‘all girls not together’.”

Why it fails: Two adjacent girls may still satisfy the question when the third girl is elsewhere.

Repair: Match BAD exactly to the negation of the required condition.

MISCONCEPTION

ONE SUBTRACTION FOR MANY BAD EVENTS

Tempting thought: “No two of A,B,C together = total - (ABC block).”

Why it fails: Pairwise adjacency can occur without all three forming one block; multiple forbidden events overlap.

Repair: Use gaps or, later, inclusion-exclusion rather than one simplistic subtraction.

METHOD CONTROL

Set logic first

Complement is not a slogan. It is a set partition. If TOTAL and BAD do not partition the universe exactly, the subtraction is invalid.

GUIDED PRACTICE

Scaffolding fades from pair complement to multiset complement

GUIDED

A | Pair nonadjacency

TRY

Fill all three lines

Seven distinct books are arranged in a row. Two specified books must not be adjacent.

TOTAL = ______
BAD = ______
GOOD = ______

B | Not all together

TRY

One forbidden group-block

Eight distinct people form a queue. Three specified people must not all be consecutive. Write the complement skeleton without evaluating.

C | Repeated letters

REASON

Rebuild both inventories

For A,A,B,B,C,D, count arrangements where C and D are not adjacent. Write the TOTAL and BAD inventories before any factorial.

D | Owner classification

REASON

Boundary control

Classify: (i) all three girls not together, (ii) no two girls together, (iii) two specified people not adjacent, (iv) circular pair not adjacent. Name the primary subtopic.

JEE TRANSFER MAP

Recognise complement through different surface forms

TRANSFER

C3/C5

Linear queue

Recognition: all girls not together

First move: define TOTAL and the exact forbidden event.

C3

Distinct word

Recognition: vowels never all together

First move: define TOTAL and the exact forbidden event.

C4

Repeated word

Recognition: specified letters do not come together

First move: define TOTAL and the exact forbidden event.

C6

Nested block

Recognition: B1/B2 nonadjacent inside boys block

First move: define TOTAL and the exact forbidden event.

DEFER circular

Circular seating

Recognition: specified pair never adjacent. Control: circular symmetry changes TOTAL and BAD, so the circular-permutation unit should own the problem.

MASTERY CHECK + LOCAL COVERAGE AUDIT

Advance only when the complement model and source slice both reconcile

AUDIT

Can you now...

1. define a common universe before subtraction?
2. translate negative wording into one positive BAD event?
3. count a forbidden adjacent pair with both internal orders?
4. distinguish “not all together” from “no two together”?
5. apply multiset division separately in TOTAL and BAD?
6. recognise when complement is only a nested substep?

Frozen ST05 corpus slice

2026	queue: all girls not together	REQUIRED
2025	DAUGHTER: vowels never all together	REQUIRED
2023	MATHEMATICS: C,S apart	REQUIRED
2021	VOWELS: consonants never all together	REQUIRED
2020	LETTER: vowels not adjacent	REQUIRED
2017	round table B1/G1 never adjacent	DEFER -> circular
2008	MISSISSIPPI no two S adjacent	DEFER -> ST06

STATUS

Audit stamp

Local ST05 slice: 5 REQUIRED, all mapped to the companion practice. Chapter-wide status remains PROVISIONAL until the complete 220-row ledger is classified.

NO TWO TOGETHER: GAP METHOD

Arrange the separators first, then place restricted objects into safe gaps

Essential Question

How do we build every valid arrangement automatically instead of subtracting many bad cases?

THIS SUBTOPIC BUILDS

1. Anchor-first gap architecture
2. Distinct / repeated restricted objects
3. Stronger spacing such as at least two separators

BOUNDARY

ST05 owns one clean forbidden event

ST06 owns repeated separation / gap placement

Circular gaps are deferred to the circular unit

HOW THIS SUBTOPIC WORKS

Build safe positions first; place the restricted group second

ST06

Concept spine

- C1 **Gap architecture** — anchor group first
- C2 **Capacity test** — m <= n+1 for basic linear gaps
- C3 **Choose gaps + arrange** — distinct vs identical specials
- C4 **Repeated anchors** — multiset base arrangements
- C5 **Stronger spacing** — at least k separators
- C6 **Gap vs complement** — pick the cleaner architecture

SCOPE **Primary owner discipline**

Linear arrangements with repeated separation conditions. Circular no-two-together questions are deferred even though they also use gaps.

Pedagogy layers

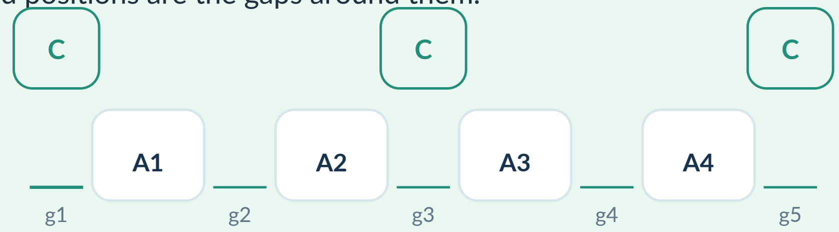
1. Real-life separation story
2. Meaning before formula
3. Gap diagram
4. Teacher model
5. Misconception repair
6. JEE trigger wording
7. Repeated-object transfer
8. Strong-spacing transfer
9. Retrieval + audit

ENTRY A

REAL LIFE

Cinema seating

Four adults are already seated in a row. Three young children must sit so that no two children are adjacent. Once the adults are placed, the safe child positions are the gaps around them.



REAL LIFE

Warning lights

Five red warning lights must be placed among eleven blue markers so that any two red lights have at least two blue markers between them. Build the compulsory blue separation first; only then place leftover blue markers.

Mandatory scaffold

Use required separators first; distribute remaining B-cubes afterward.

NOTICE

Build legality into the model

The gap method does not count invalid arrangements and then repair them. It constructs only legal positions by arranging the separating structure first.

THE CORE PICTURE: ANCHORS FIRST, GAPS SECOND

Same structure in people, letters and objects

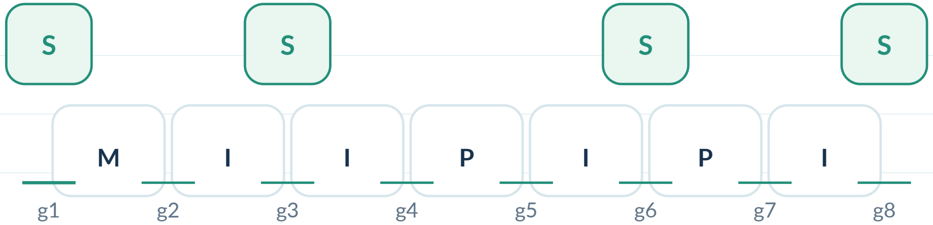
ENTRY B

People in a row

Arrange 4 boys first. They create 5 safe gaps for girls if no two girls may be adjacent.

Words with repeated letters

For MISSISSIPPI, first arrange the non-S letters. Every distinct base word creates eight gaps into which the four identical S letters can be inserted one per gap.



ABSTRACT

The n+1 fact

n anchors in a row create n+1 gaps: one before, n-1 between, and one after. Selecting different gaps keeps the restricted objects separated.

ABSTRACT

Representation survives

The same gap picture survives repeated letters. Only the count of base arrangements changes because identical anchors create multiset symmetry.

JEE TRIGGER

Language cue

Phrases like no two ... together, no two ... adjacent, separated by at least, or between any two ... should make you look for a separator group and its gaps.

C1 | THE GAP ARCHITECTURE

Arrange the unrestricted/separator group before the restricted group

C1

Teacher model: 4 boys and 3 girls

STEP 1

Place separators first

Arrange the 4 distinct boys: 4!. This arrangement becomes the anchor skeleton.

STEP 2

Safe positions

The boys create 5 gaps. Choose 3 of them for the girls, then arrange the 3 distinct girls in those selected gaps.

Why this order matters

IF GIRLS GO FIRST

Wrong construction order

Placing the restricted group first does not reveal which later boy placements will separate them. You create many illegal branches and then need repair.

IF BOYS GO FIRST

Legal by design

Every chosen gap can hold at most one girl. Separation is automatic, so the construction itself enforces the condition.

REPAIR HABIT

Anchor-first habit

Ask: Which group can act as separators? Arrange that group first. The restricted group should usually be placed second.

C2

Basic linear separation

If n anchor objects are arranged in a row, there are n+1 gaps. To place m restricted objects with no two adjacent, at most one restricted object may occupy each gap.

feasible only if m <= n + 1

EXAMPLE

Tight capacity

4 girls and 3 boys, with no two girls together:

3 boys create 4 gaps. Since 4 <= 4, the arrangement is possible - every gap must be used.

IMPOSSIBLE CASE

Capacity first

5 girls and 3 boys with no two girls together: 3 boys create only 4 gaps, but 5 girls need 5 separate gaps. Answer = 0 before any factorial.

Why students miss this

MISCONCEPTION

GAP OFF BY ONE

Tempting thought: “n anchors create n gaps.”

Why it fails: A row has one outside gap before the first anchor and one after the last anchor, in addition to the internal gaps.

Repair: Draw _ A _ A _ once. Count the underscores, not the spaces between anchors only.

RETRIEVAL

Prediction before arithmetic

Without calculating: 8 boys and 10 girls, no two girls together. Possible or

JEE TRIGGER

Fast filter

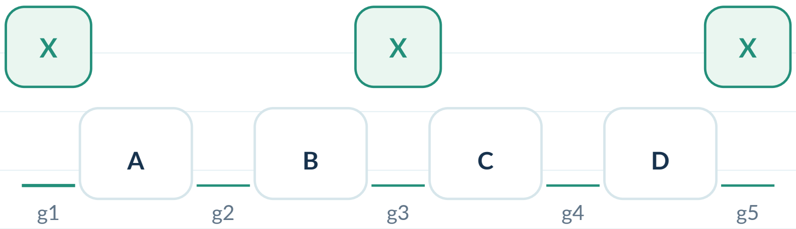
When the restricted group is almost as large as the separator group, do the capacity check before counting. It may collapse the entire problem.

C3 | CHOOSE GAPS, THEN ASK WHETHER THE SPECIAL OBJECTS ARE DISTINCT

The selected positions and internal identities are separate decisions

C3

Distinct restricted objects



MODEL

Choose then arrange

4 distinct anchors -> 5 gaps. Place 3 distinct special objects:

C(5,3) ways to choose gaps, then 3! ways to assign the three objects to those gaps.

MISCONCEPTION

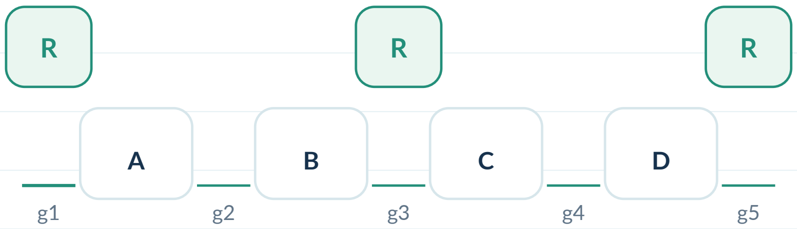
ARRANGE IDENTICAL SPECIALS

Tempting thought: “Choose the gaps and multiply by 3! anyway.”

Why it fails: Swapping identical special objects does not create a new arrangement.

Repair: Separate position choice from identity assignment.

Identical restricted objects



MODEL

Identity matters

If the 3 special objects are identical, selecting the 3 gaps completely determines their placement. There is no 3! factor.

MISCONCEPTION

FORGOT SPECIAL ARRANGEMENT

Tempting thought: “Choose 3 gaps and stop, even though the girls are distinct.”

Why it fails: Different girls in the same chosen gaps create different outcomes.

Repair: After choosing positions, ask whether identities can still vary.

C4 | REPEATED ANCHORS: THE GAP PICTURE STAYS, THE BASE COUNT CHANGES

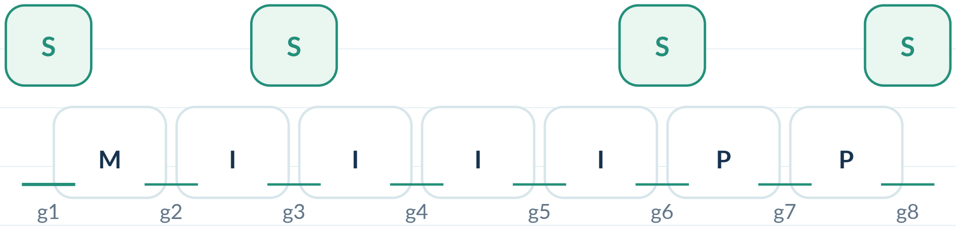
MISSISSIPPI is the canonical multiset-gap transfer

C4

Step 1: remove the restricted S letters

MISSISSIPPI has M1, I4, S4, P2. To keep S letters apart, first arrange the seven non-S letters M, I,I,I,I, P,P.

base arrangements = 7! / (4! 2!)



STEP 2

One S per selected gap

Every base word creates 8 gaps. Because the four S letters are identical, choose 4 of the 8 gaps:

C(8,4).

Why repeated anchors do not destroy gaps

REPRESENTATION

Count after arranging the base

Even when several anchor symbols look identical, each finished base arrangement still has a left edge, right edge and positions between consecutive written symbols. Those are the physical gaps.

[7!/(4!2!)] x C(8,4)

MISCONCEPTION

COUNT GAPS BEFORE THE BASE

Tempting thought: “There are repeats, so perhaps the number of gaps must also be divided.”

Why it fails: The symmetry division belongs to the number of distinct base arrangements. Each realized row still contains 8 actual insertion gaps.

Repair: Count distinct base rows first; then inspect one row for gaps.

C5 | GAP OR COMPLEMENT?

Choose the method that models the condition most directly

C5

Specified pair not adjacent

USUALLY COMPLEMENT

ST05 owner

Only one forbidden adjacency event exists. Count all arrangements, subtract the arrangements where that pair forms a block.

TOTAL - pair-together

Many restricted objects: no two adjacent

USUALLY GAPS

ST06 owner

Trying to subtract all possible adjacency patterns creates overlaps. Arrange separators first and place each restricted object in a different safe gap.

Diagnostic question

ASK

Method selection

Is there one clean bad event I can count? -> complement.

Or must I enforce repeated separation across a whole group? -> gaps.

COMPARE

Language is not enough

“A and B not adjacent” and “no two girls adjacent” sound similar in English but have different combinatorial geometry. One is one forbidden edge; the other is a global spacing condition.

C6

JEE 2022 model: red and blue cubes

There are 5 identical red cubes and 11 identical blue cubes. Between any two red cubes there must be at least 2 blue cubes.

Mandatory scaffold

R B B R B B R B B R

Use required separators first; distribute remaining B-cubes afterward.

MANDATORY BLUE **Build the minimum structure**

Five red cubes create four internal red-to-red separations. Each needs 2 blue cubes, so 8 blue cubes are compulsory.

LEFTOVERS **Residual freedom**

After using 8 blue cubes, 3 blue cubes remain. They may be distributed among the 6 regions: before first R, four internal gaps, after last R.

Residual distribution

HELPER **Gap method -> stars and bars**

Let $x_1, \ldots, x_6$ be the additional blue cubes placed in the six regions. Then

$x_1 + \ldots + x_6 = 3$, $x_i \geq 0$.

$$\text{number} = C(3+6-1, 6-1) = C(8,5)$$

MISCONCEPTION **ONE-PER-GAP ONLY**

Tempting thought: “Gap method always means choose different gaps.”

Why it fails: That works for “no two together,” but stronger spacing may leave several unrestricted separators to distribute in the same region.

Repair: Build the compulsory spacing, then count the residual freedom.

EXACTLY k BETWEEN vs AT LEAST k BETWEEN

One word changes the degrees of freedom

COMPARE

Exactly 2 separators

EXACTLY

No internal extras

If exactly two boys must lie between each pair of consecutive girls, the internal spacing is fixed. Extra boys cannot enter those internal gaps without violating "exactly."

Mandatory scaffold

R B B R B B R B B R

Use required separators first; distribute remaining B-cubes afterward.

MISCONCEPTION

AT LEAST TREATED AS EXACTLY

Tempting thought: "Place the minimum separators and stop."

Why it fails: The minimum only guarantees legality; extra separators may still be placed in many locations.

Repair: After building the minimum structure, ask what objects remain and where they may go.

At least 2 separators

AT LEAST

Residual distribution

First place two compulsory separators in every internal gap. Any remaining separator objects may be added to the ends or to those same internal regions.

Mandatory scaffold

R B B R B B R B B R

Use required separators first; distribute remaining B-cubes afterward.

MISCONCEPTION

EXACTLY TREATED AS AT LEAST

Tempting thought: "Distribute leftover separators anywhere."

Why it fails: Extra separators inside a prescribed internal gap violate the word exactly.

Repair: Underline exactly / at least before constructing the model.

C7

JEE 2025 model

Five boys and four girls sit in a row so that either all five boys sit together or no two boys sit together.

Branch B - ST06

BRANCH A - ST04 **Block architecture**

All boys together: glue the 5 boys into one block. Arrange [BBBBB] plus 4 distinct girls, then arrange the boys inside their block.

GAP **Tight-capacity gaps**

Arrange the 4 girls first: 4!. They create exactly 5 gaps. To separate all 5 boys, every gap must contain one boy, and the 5 distinct boys can be assigned in 5! ways.

$$5! \times 5!$$

$$4! \times 5!$$

DISJOINTNESS **No overlap term**

With 5 boys, the events “all boys together” and “no two boys together” cannot occur simultaneously. Therefore the OR is a direct sum of the two branch counts.

FINAL STRUCTURE **Hybrid transfer**

Required count = block branch + gap branch. The important skill is not memorizing 17280; it is recognizing that the word OR can connect two different subtopic engines.

THREE EXAMSIDE MODELS: SAME GAP FAMILY

Distinct people, identical cubes, repeated letters

MODELS

2025 row seating

MODEL

Distinct anchors + specials

No two of 5 boys together among 4 girls:

4! x 5!

This is one branch of the OR question.

OPEN 2025 PYQ

2022 cubes

MODEL

Stronger spacing

5 identical red, 11 identical blue, at least 2 blue between red:

8 blue mandatory -> distribute remaining 3 across 6 regions.

OPEN 2022 PYQ

AIEEE 2008 MISSISSIPPI

MODEL

Repeated-object gaps

Arrange M,I4,P2 first: 7!/(4!2!). This creates 8 gaps. Choose 4 gaps for the identical S letters:

[7!/(4!2!)] x C(8,4).

OPEN 2008 PYQ

COMMON DNA

Anchor -> gaps -> restricted

In every case, safe positions are manufactured before the restricted objects are placed. That is the defining gap-method move.

MISCONCEPTION CLINIC A

Three failures in basic gap construction

MIS A

MISCONCEPTION

GAP OFF BY ONE

Tempting thought: “Four anchors create four gaps.”

Why it fails: A linear row has two outside gaps in addition to the internal spaces.

Repair: Draw underscores before and after the anchor row; n anchors give n+1 gaps.

MISCONCEPTION

RESTRICTED GROUP FIRST

Tempting thought: “Arrange the girls first and then fit boys around them.”

Why it fails: The restricted group does not create the safe-position structure; invalid adjacency can reappear.

Repair: Arrange the separator/unrestricted group first.

MISCONCEPTION

FORGOT IDENTITY ASSIGNMENT

Tempting thought: “Choose three gaps for three distinct girls and stop.”

Why it fails: The selected gaps do not say which girl occupies which selected gap.

Repair: After choosing positions, ask whether the restricted objects can still permute.

REPAIR HABIT

Stable sequence

Write the three verbs before arithmetic:

ARRANGE anchors -> IDENTIFY gaps -> PLACE restricted objects.

MISCONCEPTION CLINIC B

Stronger spacing, repeats and boundary drift

MIS B

MISCONCEPTION

IDENTICALS OVERCOUNTED

Tempting thought: “Multiply by r! after choosing gaps for r identical symbols.”

Why it fails: Swapping identical symbols creates no new arrangement.

Repair: Only multiply by an assignment factorial when identities are distinct.

MISCONCEPTION

AT LEAST = EXACTLY

Tempting thought: “Use the minimum separators and stop.”

Why it fails: Extra separators may still be legally placed, so the minimum scaffold is not the whole count.

Repair: Count the residual objects after the compulsory scaffold.

MISCONCEPTION

CIRCULAR GAPS TREATED AS LINEAR

Tempting thought: “Seven anchors around a circle create eight gaps.”

Why it fails: A circle has no two ends. Seven circularly arranged anchors create seven cyclic gaps.

Repair: Defer circular symmetry and cyclic gaps to the circular-permutation unit.

BOUNDARY CHECK

Prevent subtopic drift

Before using n+1, ask: Is the arrangement linear? If the problem is circular, both the anchor count and gap count change.

GUIDED PRACTICE

Scaffolding fades from basic gaps to stronger spacing

GUIDED

A | Basic gap count

TRY

Fill the skeleton

Arrange 6 boys and 4 girls in a row so that no two girls are together.

boys: ______
gaps: ______
choose gaps: ______
arrange girls: ______

B | Identical restricted objects

TRY

Identity reasoning

A row contains 7 distinct books and 3 identical markers, with no two markers adjacent. Write a counting expression and explain why no 3! appears.

C | Stronger spacing

REASON

Do not finish arithmetic

Five identical R objects and 12 identical B objects are placed in a row with at least 2 B between consecutive R. How many B are compulsory? How many regions receive the leftovers?

D | Method owner

CLASSIFY

ST05 / ST06 / ST04 / circular

Choose the primary engine: (i) A and B not adjacent, (ii) no two girls adjacent, (iii) all boys together, (iv) no two girls adjacent around a circle.

MASTERY + LOCAL EXAMSIDE AUDIT

A learner checklist and a corpus checklist

AUDIT

You should now be able to...

1. choose the separator/anchor group
2. draw n+1 linear gaps
3. run the capacity test before counting
4. separate gap choice from identity assignment
5. handle repeated anchors without changing gap geometry
6. build at least-k spacing from compulsory separators
7. distinguish exactly from at least
8. choose gap vs complement by event geometry
9. combine a gap branch with an OR/block branch

Frozen ST06 ExamSIDE slice

REQUIRED — **2025 boys/girls OR: gap branch**
ST06.C7

REQUIRED — **2022 red/blue cubes, at least 2 blue**
ST06.C6

REQUIRED — **AIEEE 2008 MISSISSIPPI, no two S**
ST06.C4

DEFER — **2023 round-table no two girls**
Circular unit

DEFER — **AIEEE 2003 round-table no two women**
Circular unit

LOCAL STATUS — **Release reconciliation**

Required: 3 | Included in companion: 3/3
Source links: 3/3 | Concept links: 3/3 | Solutions: 3/3
H-policy failures: 0 | Readability failures: 0

Chapter status: PROVISIONAL until the full 220-question ledger is frozen.

NUMBER FORMATION

Build the number from restricted positions - never count digits as if every slot were identical

1000s	100s	10s	1s
5	?	?	0

Essential Question

Which positions are special before the ordinary permutation engine begins?

THIS SUBTOPIC BUILDS

1. Leading-zero discipline
2. Range / parity / divisibility case design
3. Repetition and at-least-once control

BOUNDARY

Pure word arrangements remain ST01-ST06

Dictionary order is ST08

Inclusion-exclusion appears only when digit usage forces it

HOW NUMBER-FORMATION QUESTIONS ARE BUILT

Read the special positions before multiplying

ST07

Concept spine

C1	**Leading digit**	zero cannot lead
C2	**Reuse rule**	same pool vs shrinking
C3	**Range**	prefix / first-digit restriction
C4	**Parity / 5 / 10**	units digit controls
C5	**Divisibility 3 / 9**	digit sum controls
C6	**Combined divisibility**	two constraints at once
C7	**Divisibility by 11**	alternating-place sum
C8	**At least once**	surjection / multiplicity cases

Decision order

1. LOCK SPECIAL SLOTS — **Restrictions first**

First digit? Last digit? Fixed range prefix? Divisibility-controlled positions?

2. COUNT THE REMAINING POOL — **Then ordinary counting**

Only after the special positions are fixed should you decide whether choices repeat or shrink.

3. SPLIT CASES ONLY WHEN POOLS DIFFER — **Case architecture**

Casework is justified when a choice changes the legal pool for a later slot - especially zero at the units place versus a nonzero even digit.

START WITH THE DIFFERENCE BETWEEN A CODE AND A NUMBER

A leading zero is harmless in a code, but destroys a digit in a number

ENTRY

REAL LIFE

Locker code 0427

A four-symbol code may begin with 0. All four positions are simply labelled slots. If repetition is allowed, every slot may have the full symbol pool.

1	2	3	4
0	4	2	7

REAL LIFE

Four-digit number 0427?

As a number, 0427 is just 427. Therefore the thousands place cannot be 0 when the question asks for a four-digit number.

1000s	100s	10s	1s
~~0~~	4	2	7

NOTICE

Meaning before notation

The same symbols can represent different counting objects. Before using a formula, decide whether the leftmost position has a semantic restriction.

C1 | ZERO MAKES THE FIRST SLOT SPECIAL

Treat the leading position before the other slots

C1

Without repetition

Form a 5-digit number from {0,1,2,3,4,5} without repetition. The first slot has only five legal choices: 1-5. After that, the remaining pool contains five symbols, including 0.

1st	2nd	3rd	4th	5th
5	5	4	3	2

$$5 \times 5 \times 4 \times 3 \times 2$$

With repetition

If repetition is allowed, the first slot still cannot be 0, but every later slot regains the full six-digit pool.

1st	2nd	3rd	4th	5th
5	6	6	6	6

$$5 \times 6^4$$

MISCONCEPTION

ZERO TREATED LIKE EVERY OTHER DIGIT

Tempting thought: “There are 6 choices for each of 5 slots.”

Why it fails: That count includes strings beginning with 0, which are not five-digit numbers.

Repair: Separate the leading slot before applying a uniform power or permutation.

JEE TRIGGER

Read the first digit first

Words such as n-digit integer, greater than 1000, or between 5000 and 9000 immediately make the leading position special.

C2 | REPETITION: DOES THE DIGIT POOL RESET?

Position labels and digit identities are different issues

C2

Reuse allowed

1	2	3	4
4	4	4	4

Reuse forbidden

1	2	3	4
4	3	2	1

MODEL **Pool resets**

Using digits {1,3,5,8} to form 4-digit strings with repetition allowed: every slot sees the same four symbols.

MODEL **Pool shrinks**

Without repetition, every used digit disappears from the later pool.

MISCONCEPTION **DIFFERENT SLOTS = DIFFERENT DIGITS**

Tempting thought: “Because the positions are different, the digits must also be different.”

Why it fails: Distinct positions do not imply a no-repetition rule.

Repair: Read the reuse sentence separately from the position structure.

MISCONCEPTION **POWER USED BY HABIT**

Tempting thought: “There are k symbols, so use k^n.”

Why it fails: A power assumes every slot has the same pool. Leading zero, no-repetition and divisibility constraints can break that symmetry.

Repair: Write the legal pool over each slot before compressing.

C3

Between 5000 and 10000

Using {1,3,5,7,9} without repetition, any valid number in this interval must be 4-digit and begin with 5, 7 or 9.

1000s	100s	10s	1s
3	4	3	2

MODEL **Prefix controls the interval**

Choose the thousands digit first: 3 choices. Then arrange any 3 of the remaining 4 digits in the last three places.

$$3 \times 4P3 = 72$$

General prefix rule

HELPER **Lexicographic thinking for numbers**

For inequalities, compare from the left. If the first digit already places the number safely inside the interval, later positions return to ordinary counting. If it ties the boundary prefix, continue to the next digit.

REAL LIFE **Prefix comparison**

This is the same logic used to sort invoice numbers or timestamps: the earliest position at which two strings differ determines the order.

MISCONCEPTION **COUNT ALL THEN GUESS A FRACTION**

Tempting thought: “Roughly half the permutations should exceed the boundary.”

Why it fails: Digit sets and boundary prefixes are rarely symmetric enough for such fractions.

Repair: Lock the earliest restricted digit explicitly.

C4

Even number with zero present

For {0,1,3,4,6,7} without repetition, an even number ends in 0, 4 or 6. But the later pool differs when the last digit is 0 versus 4/6 because zero cannot lead.

CASE A: units = 0 — **Zero used safely**

Hundreds place: 5 nonzero choices. Tens place: 4 remaining. Count = 5 x 4.

CASE B: units = 4 or 6 — **Zero remains in pool**

2 choices for the units digit. Hundreds place has 4 nonzero choices; tens place then 4 choices. Count = 2 x 4 x 4.

Divisible by 5

CONTROL SLOT — **Do the units slot first**

Units digit must be 0 or 5. If both are available, usually split into two cases because choosing 0 versus 5 changes what is legal in the leading slot.

even example: 20 + 32 = 52

MISCONCEPTION — **ONE CASE FOR ALL EVEN LAST DIGITS**

Tempting thought: “There are 3 even last digits, so multiply one common count by 3.”

Why it fails: The leading-zero restriction makes the remaining pools different when 0 is used versus when 0 remains available.

Repair: Split whenever the special choice changes a later legal pool.

C5

Digit-sum filter

RULE

Selection filter before arrangement

A number is divisible by 3 when its digit sum is divisible by 3; divisible by 9 when its digit sum is divisible by 9. The property depends on the chosen digits, not their order.

WORKFLOW

Filter -> arrange

1. Determine which digit selections/repetition patterns have the required sum residue.
2. For each legal selection, count its distinct arrangements.
3. Repair the leading-zero cases if 0 is included.

MISCONCEPTION

TEST EVERY PERMUTATION

Tempting thought: “Generate arrangements and check divisibility one by one.”

Why it fails: Digit sum is invariant under permutation, so all arrangements of one chosen digit multiset share the same divisibility-by-3/9 status.

Repair: Move the divisibility test before arrangement.

Residue helper

H2 VISUAL

Residue classes

Group available digits by residue mod 3: 0, 1, 2. For repetition-allowed questions, count residue-pattern sequences before expanding to actual digits.

Example digits {0,1,2,5,9}: residue classes have sizes 2, 1, 2. If the first digit is fixed at 5 (residue 2), the final three digits must sum to residue 1 mod 3.

JEE TRIGGER

Think modulo before cases

When repetition is allowed and the digit set is small, residue classes can be faster and less error-prone than listing numerical cases.

C6 | COMBINE LOCAL AND GLOBAL DIVISIBILITY CONDITIONS

Example: multiples of 6 and 15

C6

Multiple of 6

TWO TESTS

Local slot + global sum

Divisible by 6 means divisible by 2 and 3. One condition controls the units digit; the other controls the chosen digit sum.

ORDER OF ATTACK

Choose the cleaner direction

Filter digit selections by sum mod 3, then count arrangements ending in an even digit. Or, if the even choices are few, fix the units digit first and test the remaining pool.

Multiple of 15

TWO TESTS

Forced special slot

Divisible by 15 means divisible by 5 and 3. If 0 is absent and 5 is present, the units digit may be forced to 5, greatly simplifying the problem.

JEE 2022: units = 5; choose 5 of remaining 6 with sum condition

MISCONCEPTION

MULTIPLY INDEPENDENT FRACTIONS

Tempting thought: “Half are even and one-third are divisible by 3, so take one-sixth.”

Why it fails: The two properties need not be independent over a restricted digit set without repetition.

Repair: Count the legal structures directly.

C7

Alternating-sum rule

RULE **Group positions, not just digits**

For a number to be divisible by 11, the difference between the sum of digits in alternating positions must be a multiple of 11 (including 0).

1	2	3	4	5	6
+	-	+	-	+	-

WORKFLOW **Select groups -> arrange**

Choose which digits occupy the odd-position group so that the alternating-sum condition works. Then arrange within odd and even positions, repairing a leading-zero case if needed.

Misconception

MISCONCEPTION **CHECK LAST TWO DIGITS**

Tempting thought: “Divisibility rules always depend on the final digit(s).”

Why it fails: That is true for 2, 4, 5, 8, 10, etc., but 11 is a positional alternating-sum rule across the whole number.

Repair: Write + - + - signs over the positions before selecting digits.

TRANSFER **Future audit row**

The JEE 2019 six-digit problem using 0,1,2,5,7,9 is a natural ST07 owner: the alternating-position groups determine which digit partitions are legal, then permutations finish the count.

C8 | EVERY ALLOWED DIGIT MUST APPEAR

At least once: use missing-digit complement or multiplicity patterns

C8

Method A: inclusion-exclusion

Seven positions are filled from five nonzero digits, and every digit must appear. Start from all 5^7 strings and subtract strings missing at least one required digit.

$$5^7 - C(5,1)4^7 + C(5,2)3^7 - C(5,3)2^7 + C(5,4)$$

WHY IT WORKS **Onto-count structure**

Missing-digit events overlap. Inclusion-exclusion corrects the over-subtraction systematically.

MISCONCEPTION **SUBTRACT EACH MISSING DIGIT ONCE**

Tempting thought: “Use 5^7 - 5 x 4^7 and stop.”

Why it fails: Strings missing two digits were subtracted twice, so overlaps must be restored.

Repair: Whenever several forbidden missing-symbol events can occur together, ask for overlap correction.

Method B: multiplicity patterns

PATTERNS **ST02 transfer**

With 7 positions and 5 required digits, the extra two occurrences can be distributed as either 3,1,1,1,1 or 2,2,1,1,1. Count each multiset pattern and add.

METHOD CHOICE **Two representations**

Inclusion-exclusion scales cleanly when the alphabet is small. Multiplicity cases may be more intuitive when only a few extra repetitions exist. Both should agree.

JEE MODELS: READ THE CONTROL SLOT FIRST

Three recurring number-formation fingerprints

MODEL

MODEL 1 | EVEN NUMBER Case split driven by zero

Question gives 0 plus several nonzero digits without repetition.

Decision: split units = 0 versus units = nonzero even because the leading pool changes.

MODEL 3 | RANGE + DIVISIBLE BY 3 Prefix + modulo

Question fixes the thousands digit indirectly through the interval, allows repetition, and adds a digit-sum condition.

Decision: fix the prefix; count residue patterns in the remaining slots.

MODEL 2 | RANGE + DIVISIBLE BY 5 Two special slots

Question asks for five-digit numbers above a threshold and divisible by 5.

Decision: units is 0/5; leading digit must satisfy the range; then permute the middle positions.

MODEL 4 | EVERY DIGIT USED Surjection / multiset transfer

Seven positions from five allowed digits, each required at least once.

Decision: use inclusion-exclusion or the two multiplicity patterns; do not treat it as 5^7.

M1

MISCONCEPTION

ZERO LEADING

Tempting thought: “The first slot has the same choices as the others.”

Why it fails: A leading 0 shortens the number, so it is not an n-digit outcome.

Repair: Mark the first slot before any multiplication.

MISCONCEPTION

UNEQUAL CASES MERGED

Tempting thought: “There are three valid last digits, so multiply one common count by 3.”

Why it fails: If 0 is one of those last digits, the later leading-slot pool differs from the nonzero-last-digit cases.

Repair: Check whether each case leaves the same remaining legal pool.

MISCONCEPTION

RANGE COUNTED AFTERWARD

Tempting thought: “Count all permutations, then estimate which lie in the interval.”

Why it fails: Thresholds act lexicographically from the first digit; there may be no symmetric fraction.

Repair: Constrain the earliest decisive digit first.

SELF-CHECK

Diagnose the slot

For each wrong model, name the exact slot whose legal pool was misrepresented.

MISCONCEPTION CLINIC II

Divisibility and overlap errors

M2

MISCONCEPTION

DIGIT-SUM TEST USED AFTER ARRANGING

Tempting thought: “List permutations and then check divisibility by 3.”

Why it fails: The digit sum is unchanged by rearrangement, so the test belongs at the digit-selection stage.

Repair: Filter the digit multiset first, then arrange.

MISCONCEPTION

DIVISIBILITY CONDITIONS TREATED INDEPENDE

Tempting thought: “One-half even and one-third divisible by 3, so one-sixth are divisible by 6.”

Why it fails: Restricted digit pools create dependence between the last digit and the available digit sum.

Repair: Count legal cases directly rather than multiplying probability-like fractions.

MISCONCEPTION

MISSING IE OVERLAP

Tempting thought: “Subtract strings missing each digit and stop.”

Why it fails: A string can miss multiple digits, so bad events overlap.

Repair: Use inclusion-exclusion or a multiplicity-pattern count.

REPAIR HABIT

Structural diagnosis

Before counting, label each condition as position-local, digit-set global, or overlap condition. That classification determines the architecture.

GUIDED PRACTICE + TRANSFER

Scaffold the special positions, then remove the scaffolding

GP

A. Complete

1 3-digit even numbers from {0,2,3,5} without repetition

Units cases: 0 or 2 -> ______

2 4-digit numbers > 6000 from {1,4,6,7,9} without repetition

Thousands choices: ______

3 5-digit multiples of 5 from {0,1,2,5,7,8} without repetition

Units cases:

EXPLAIN **Explain the pool change**

Why does choosing units = 0 often require a separate case from units = 4 or 6?

B. JEE fingerprints

2021 — **3-digit even; 0 present; no repetition** — C1/C4

2022 — **6-digit multiple of 15; no repetition** — C5/C6

2023 — **>40000 and divisible by 5** — C3/C4

2026 — **5000<n<9000, divisible by 3, repetition** — C3/C5

2026 — **7 digits; five symbols each at least once**

TRANSFER **Owner question**

When multiple restrictions appear, do not ask “which formula?” Ask “which slot is controlled first, and what does that choice do to the remaining pool?”

MASTERY + LOCAL COVERAGE AUDIT

Can you classify the restriction before counting?

MASTER

You should now be able to...

1. separate codes from n-digit numbers
2. protect the leading digit from zero
3. distinguish repetition allowed/forbidden
4. turn inequalities into prefix restrictions
5. fix parity / divisibility-by-5 units slots
6. use digit sums for 3 and 9
7. combine divisibility conditions structurally
8. use alternating-position groups for 11
9. count every-symbol-at-least-once strings

Frozen ST07 ExamSIDE slice

REQUIRED **5 questions**

Q1 2021 even number
Q2 2023 >40000 and divisible by 5
Q3 2022 multiple of 15
Q4 2026 interval + divisible by 3
Q5 2026 every digit at least once

DEFER **One primary owner**

Dictionary serial-order questions -> ST08.
Repeated-digit position problems whose primary engine is multiset arrangement -> ST02/ST03.

OPEN EXAMSIDE CHAPTER

DICTIONARY / LEXICOGRAPHIC RANK

Count complete prefix blocks before walking into the target word

A...	120 words in this prefix bucket
D...	120 words in this prefix bucket
M...	target words in this prefix bucket

Essential Question

How many complete dictionary blocks lie before the target prefix?

THIS SUBTOPIC BUILDS

1. Prefix-bucket counting
2. Rank with repeated letters
3. Inverse problem: find the kth word

BOUNDARY

Ordinary word arrangements -> ST01/ST02

Alphabetical-order constraints without rank -> positional/relative-order units

This unit owns serial / rank / kth-word questions

THE PREFIX-BUCKET METHOD

Dictionary order is repeated counting of complete suffix sets

ST08

Concept spine

C1	**Lexicographic order**	earliest differing letter
C2	**Distinct-letter rank**	count smaller prefixes
C3	**Repeated letters**	multiset suffix blocks
C4	**Rank table**	prefix-by-prefix accumulation
C5	**Inverse kth word**	locate bucket, subtract, repeat
C6	**Off-by-one control**	before + 1
C7	**Embedded rank**	probability/sequence uses rank

One sentence to remember

MASTER IDEA **Complete blocks before target**

At each position, count all valid words formed by placing a smaller available letter there while keeping the already-matched prefix fixed. Then move to the next target letter.

REPRESENTATION **Prefix tree**

Think of the dictionary as nested folders: first-letter folders contain second-letter folders, and each folder contains all valid suffix permutations.

STOP RULE **Off-by-one control**

When every position matches the target, add 1 for the target word itself. Rank = words before + 1.

START WITH SORTING, NOT WITH FACTORIALS

Lexicographic order is how filenames and IDs are grouped

ENTRY

REAL LIFE

Sorted filenames

Imagine all 5-letter filenames made from A,B,C,D,E. Every filename beginning with A appears before every filename beginning with B. The dictionary is therefore built from large prefix blocks.

A...	24 words in this prefix bucket
B...	24 words in this prefix bucket
C...	24 words in this prefix bucket

REAL LIFE

Phonebook search

To find THAMS among permutations of MATHS, you do not list 120 words. You skip whole groups: all A..., H..., M..., S... words, then enter the T... group.

A...	24 words in this prefix bucket
H...	24 words in this prefix bucket
M...	24 words in this prefix bucket
S...	24 words in this prefix bucket

NOTICE

Count blocks, not words

Dictionary rank is a search algorithm: skip complete blocks whenever their prefix is certainly smaller than the target prefix.

C1 | EARLIEST DIFFERING LETTER DECIDES THE ORDER

Lexicographic comparison is local from the left

C1

Compare two words

EXAMPLE

Left-to-right dominance

Compare MONDAY and MONEYD. M matches M, O matches O, then N is compared with N; the first differing position decides which comes first. Later letters cannot reverse that decision.

WHY PREFIXES WORK

Complete block

Once a prefix is smaller than the target prefix at the first differing position, every suffix attached to that prefix lies before the target.

Rank logic

STEP 1

Alphabetical inventory

Sort the available letters alphabetically.

STEP 2

Skip buckets

At the current position, count unused letters smaller than the target letter. For each such choice, count all distinct arrangements of the remaining suffix.

STEP 3

Walk into the target

Fix the actual target letter and move one position right. Repeat until the full word is matched; then add 1.

C2

Model: MONDAY

Alphabetical order: A, D, M, N, O, Y. For target MONDAY, first count first letters smaller than M: A and D.

A...	120 words in this prefix bucket
D...	120 words in this prefix bucket

$2 \times 5! = 240$

CONTINUE **Prefix accumulation**

Now fix M. At the second position, letters smaller than target O among the remaining {A,D,N,O,Y} are A,D,N: each contributes 4!. Continue position by position.

Compact rank table

pos	target	smaller available	block size
1	M	A,D	5! each
3			

OFF-BY-ONE **Rank = before + 1**

The accumulated total counts words before MONDAY. Add 1 for MONDAY itself.

C3 | REPEATED LETTERS CHANGE THE BUCKET SIZE

Every suffix block is a multiset permutation count

C3

Model: UDAYPUR

The letters are A, D, P, R, U, U, Y. Because U repeats twice, a skipped first-letter bucket does not contain 6! distinct suffixes when both U copies

PREFIX A...

Multiset suffix

Fix A. Remaining letters D,P,R,U,U,Y contain two identical U letters, so the bucket size is $6!/2!$, not $6!$.

bucket size = 360

UPDATE AFTER EACH FIX

Recompute the suffix inventory

The repeated-letter denominator changes as copies are consumed. Once one U is fixed, the remaining suffix may contain only one U and become an ordinary factorial.

Misconception

MISCONCEPTION

USE THE SAME FACTORIAL EVERYWHERE

Tempting thought: “Every first-letter bucket contains $(n-1)!$ words.”

Why it fails: That assumes all remaining letters are distinct. Repeated letters collapse permutations inside a bucket.

Repair: At every prefix step, write the multiset of letters still available and recompute its distinct-permutation count.

JEE TRIGGER

Same algorithm, new block size

Words like QUEEN, SMALL, BHBJO, UDAYPUR contain repeated letters. The rank algorithm is unchanged; only the suffix-block formula changes.

C4 | USE A RANK TABLE TO PREVENT SKIPPED OR DOUBLE-COUNTED BLOCKS

Make every prefix decision auditable

C4

Recommended table

Position	Target	Smaller unused	Suffix count	Running tot
1	T	A,H,M,S	4 x 4!	96
2	H	A	1 x 3!	102
3	A	-	0	102
4	M	-	0	102
5	S	-	0	102

MODEL: THAMS

Auditable calculation

The table accumulates 102 words before THAMS, so THAMS has rank 103.

Why the table helps

REASON 1

No reused letters

It forces you to update the available letters after fixing each target prefix.

REASON 2

Two-factor view

It separates “number of smaller choices” from “size of each suffix bucket,” reducing mental arithmetic errors.

REASON 3

Audit every block

For repeated letters, the suffix-count column makes the changing multiset denominator explicit.

C5 | INVERSE PROBLEM: FIND THE kth WORD

Locate the bucket containing k, subtract earlier buckets, repeat

C5

Forward rank vs inverse rank

FORWARD **Rank**

Target word given -> count all earlier buckets.

INVERSE **kth word**

Serial number k given -> find which prefix bucket contains k, subtract the sizes of earlier buckets, and recurse on the suffix.

new k = old k - skipped bucket totals

Bucket location

A...	24 words in this prefix bucket
B...	24 words in this prefix bucket
C...	24 words in this prefix bucket
D...	24 words in this prefix bucket
E...	24 words in this prefix bucket

EXAMPLE **Subtract completed buckets**

For 5 distinct letters, positions 49-72 lie in the third first-letter bucket. Therefore the 50th word begins with the third letter, and its within-bucket rank is 50-48 = 2.

C6

Model: BHBJO

Letters sorted: B,B,H,J,O. There are 60 distinct words. First-letter buckets are not equal because fixing B consumes one copy of the repeated letter.

B...	24 words in this prefix bucket
H...	12 words in this prefix bucket
J...	12 words in this prefix bucket
O...	12 words in this prefix bucket

LOCATE 50

Unequal prefix buckets

B... covers ranks 1-24; H... covers 25-36; J... covers 37-48. Therefore rank 50 lies in the O... bucket with within-bucket rank 2. Continue using remaining multiset B,B,H,J.

Misconception

MISCONCEPTION

ASSUME EQUAL BUCKETS

Tempting thought: “60 words over 4 possible first letters, so 15 per bucket.”

Why it fails: Repeated-letter symmetry changes after different first letters are fixed. Fixing B removes one copy; fixing H leaves both B copies.

Repair: Compute each candidate prefix bucket from the remaining multiset.

RESULT CHECK

Inverse rank

The 50th word is OBBJH. The point is not the answer; it is the iterative bucket-location process.

C7 | “WORDS BEFORE” IS NOT THE SAME AS “RANK”

Control the final +1 deliberately

C7

Three common phrasings

A **No +1**

“How many words appear before COCHIN?” -> stop at the count before.

B **Add target**

“What is the rank / serial number / position of PUBLIC?” -> words before + 1.

C **Different direction**

“What is the 50th word?” -> inverse problem; k is already a rank, so do not add 1 during bucket subtraction.

Misconception

MISCONCEPTION **OFF-BY-ONE**

Tempting thought: “The running total is the rank.”

Why it fails: The running total usually counts complete words strictly before the target; the target itself occupies the next serial number.

Repair: Write “before = ___, rank = before + 1” as the final line.

LANGUAGE TRIGGERS **Read the noun**

rank, position, serial number -> +1 after forward counting.

before -> no +1.

kth word -> inverse bucket search.

EMBEDDED RANK: THE DICTIONARY STEP MAY BE ONLY ONE PART OF THE PROBLEM

Do the lexicographic computation first, then hand off

TRANSFER

JEE 2025 fingerprint

HYBRID

Rank is an intermediate variable

A probability sequence is indexed by dictionary-ordered words A,B,C,D,E. The probability of CDBEA depends on its serial number n.

DECISION

Separate engines

First compute the rank of CDBEA using ST08. Only after n is known should the probability recurrence be evaluated.

Owner discipline

PRIMARY OWNER

Primary engine

If the difficult recognition step is finding the lexicographic serial number, ST08 owns the question even when another chapter formula is used afterward.

AUDIT NOTE

No coverage inflation

Hybrid questions may be revisited in another unit, but they are counted once in the chapter ledger.

OPEN EXAMSIDE CHAPTER

MISCONCEPTION CLINIC I

Prefix and suffix-count failures

M1

MISCONCEPTION

DICTIONARY_PREFIX_MISCOUNT

Tempting thought: “Count only the smaller first letters and stop.”

Why it fails: After matching the first target letter, smaller choices can still occur at every later position.

Repair: Repeat the same smaller-letter test after fixing each matched prefix.

MISCONCEPTION

WRONG SUFFIX SIZE

Tempting thought: “Every skipped prefix contributes one factorial.”

Why it fails: Repeated letters in the remaining suffix reduce the number of distinct arrangements.

Repair: Recompute the multiset suffix count after every tentative prefix.

MISCONCEPTION

LETTER REUSED

Tempting thought: “At position 3, compare with all original letters again.”

Why it fails: Earlier target letters have already been consumed and are no longer available unless there are repeated copies.

Repair: Maintain an explicit remaining-letter inventory.

SELF-CHECK

Auditable terms

For any rank calculation, you should be able to point to each added term and name the exact prefix block it represents.

MISCONCEPTION CLINIC II + GUIDED PRACTICE

Rank direction and bucket location

M2 / GP

MISCONCEPTION

OFF-BY-ONE

Tempting thought: “Words before = rank.”

Why it fails: Rank includes the target word itself.

Repair: Write the final +1 as a separate line.

MISCONCEPTION

INVERSE SEARCH WITHOUT SUBTRACTION

Tempting thought: “Choose the bucket containing k, but keep the same k.”

Why it fails: Once earlier buckets are skipped, the target rank inside the chosen bucket is smaller.

Repair: Replace k by k - skipped total at every step.

Guided try

Letters A,B,C,D. Find the rank of CBAD.
First-letter blocks before C: A..., B... -> 2 x 3! = ____.
Inside C..., letters before B among A,B,D are A -> ____ x 2! = ____.
Finish and add 1.

Inverse try

Letters A,B,C,D. Find the 15th word.
Each first-letter bucket has 3! = 6 words.
Ranks 1-6: A..., 7-12: B..., 13-18: C....
So first letter = ____, new within-bucket rank = ____.

JEE TRANSFER MAP

Representative ExamSIDE dictionary fingerprints

PYQ

2023 MONDAY

C2/C4

distinct forward rank

OPEN EXAMSIDE

2023 THAMS

C2/C4

distinct forward rank

OPEN EXAMSIDE

2023 PUBLIC

C2/C4

distinct forward rank

2024 BHBJO

C5/C6

inverse kth word + repeat

2026 UDAYPUR

C3/C4

forward rank + repeated U

OPEN EXAMSIDE

PATTERN

What to recognize

JEE repeatedly changes the surface word but preserves the same prefix-bucket engine. Repeated letters and inverse rank are the main escalation points.

MASTERY + LOCAL COVERAGE AUDIT

Can you rank without listing?

MASTER

You should now be able to...

1. compare words lexicographically
2. count complete smaller-prefix buckets
3. compute distinct-letter rank
4. recompute repeated-letter suffix blocks
5. use a running rank table
6. solve kth-word inverse problems
7. control words-before vs rank +1
8. recognize rank as an embedded subproblem

Frozen ST08 ExamSIDE slice

REQUIRED **5 questions**

MONDAY | THAMS | PUBLIC | BHBJO 50th word | UDAYPUR

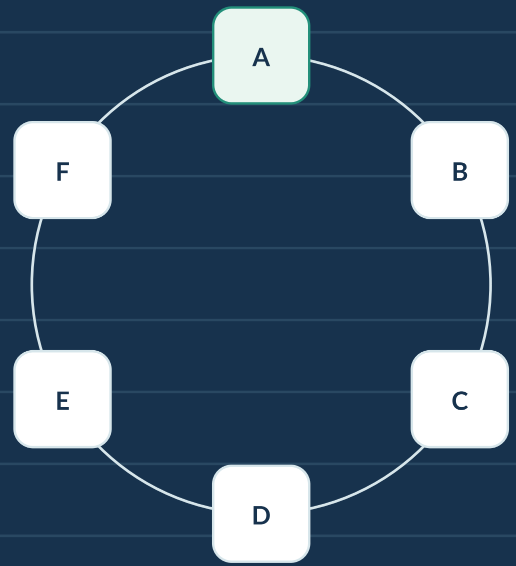
DEFER **Primary owner**

Alphabetical-relative-order questions such as “vowels in alphabetical order” are not rank problems; they remain with positional/relative-order counting.

OPEN EXAMSIDE CHAPTER

CIRCULAR PERMUTATIONS

Remove rotational duplicates first; then import blocks, complements and gaps carefully



Essential Question

When does turning the whole arrangement fail to create a new outcome?

THIS SUBTOPIC BUILDS

1. Rotation equivalence and anchoring
2. Circular block / complement / gap
3. Distinguish seats, tables and necklaces

BOUNDARY

Linear arrangements -> earlier units

Circular no-two-together returns here

Reflection equivalence belongs only when explicitly stated

CIRCULAR DECISION MAP

First decide what symmetries identify two arrangements

ST09

Concept spine

C1	**Rotation equivalence**	anchor one distinct person
C2	**Basic circle**	(n-1)!
C3	**Reflection**	usually distinct at a table
C4	**Together**	circular block
C5	**Not adjacent**	circular complement
C6	**No two together**	n cyclic gaps
C7	**Alternation**	anchor one group, fill gaps
C8	**Labelled seats**	back to n!

Three questions before counting

1 Seat identity

Are the positions physically labelled? If yes, rotations may be distinguishable.

2 Remove rotation

If the whole table rotates, is the seating considered unchanged? If yes, fix one distinct person as an anchor.

3 Reflection check

Does mirror reversal count as the same arrangement? For ordinary round-table seating, usually no.

START WITH A DINNER TABLE

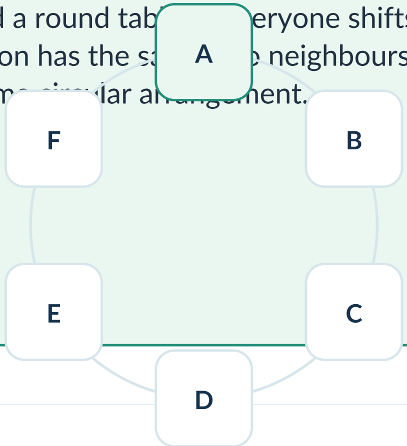
A rotated seating may describe the same neighbour relations

ENTRY

REAL LIFE

Unlabelled round table

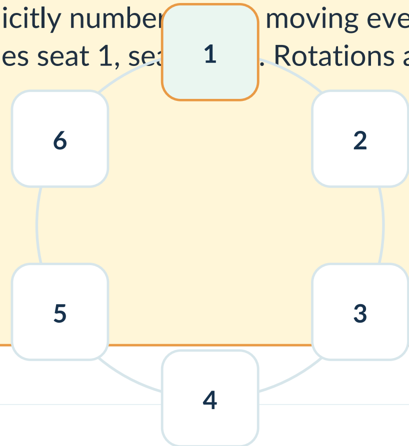
Six friends sit around a round tab eryone shifts one chair clockwise, each person has the s e neighbours. We normally regard this as the same circular arrangement.



REAL LIFE

Numbered seats

If the chairs are explicitly number moving everyone one seat changes who occupies seat 1, se . Rotations are now different outcomes.



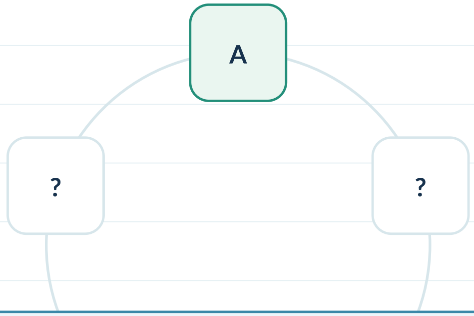
NOTICE

Meaning before notation

The formula is determined by the equivalence relation in the wording, not merely by drawing a circle.

C1

Anchor one distinct person



MODEL

Choose a representative

Fix A at a reference position. Every circular seating can be rotated so that A occupies that position, and exactly one rotation does so. Now arrange the other n-1 people around A.

Compression

$$n! / n = (n-1)!$$

WHY DIVIDE BY n?

Orbit size n

A linear listing around the circle has n possible starting points. Those n rotations describe the same circular neighbour pattern.

MISCONCEPTION

USE n! AROUND A ROUND TABLE

Tempting thought: “There are n people, so n!.”

Why it fails: That counts every circular seating once for each possible starting point.

Repair: Fix one distinct person first, then arrange the rest.

C2/C3 | ROTATION EQUIVALENT; REFLECTION USUALLY NOT

Clockwise and anticlockwise neighbour order can be different

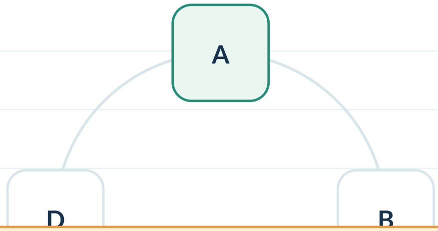
C2/C3

Ordinary seating

TABLE

Do not divide by 2

With A fixed, the clockwise sequence A-B-C-D and A-D-C-B are mirror images. At a dinner table they usually count as different because B is on a different side of A.



RULE

Reflection distinct

For ordinary circular permutations of distinct people: (n-1)!, not (n-1)!/2.

Necklace / bracelet

ONLY IF STATED

Do not import necklace rules

If an object can be flipped over and mirror images are explicitly regarded as identical, then reflection symmetry must also be factored out. That is a different equivalence problem.

MISCONCEPTION

DIVIDE BY 2 AUTOMATICALLY

Tempting thought: “Circle means rotations and reflections are the same.”

Why it fails: A table cannot usually be flipped through the plane while preserving left/right neighbours.

Repair: Ask what physical transformations the problem declares equivalent.

C4 | TOGETHER AROUND A CIRCLE

The block reduces the number of circular units

C4

Specified pair together

For n distinct people around an unlabelled round table, force A and B together. Treat [AB] as one unit along with the other n-2 people: there are n-1 circular units.

$$\text{outer} = (n-2)!$$

$$\text{inner} = 2!$$

$$\text{together} = 2(n-2)!$$

REPRESENTATION

Circular block

Glue A-B before applying circular symmetry. Do not arrange n-1 units linearly and then forget that the outer arrangement is still circular.

Block with a larger group

GENERAL

Import ST04 carefully

A block of k distinct people becomes one circular unit. Outer units = n-k+1, so outer arrangements = (n-k)!. Multiply by k! internal block orders, unless there are additional internal restrictions.

MISCONCEPTION

LINEAR OUTER COUNT

Tempting thought: “Block gives (n-k+1)! outside.”

Why it fails: The outer objects still sit on a circle, so rotational duplicates remain.

Repair: After gluing, ask again: row or circle?

C5 | SPECIFIED PAIR NOT ADJACENT

Total circular arrangements minus circular block arrangements

C5

Model

GOOD = TOTAL - BAD

TOTAL **Circular total**

n distinct people around the table -> (n-1)!

BAD **Circular block**

A and B together -> 2(n-2)!

good = (n-1)! - 2(n-2)!

JEE 2017 fingerprint

Five boys and three girls sit at a round table; specified B1 and G1 are never adjacent.

STRUCTURE **Complement in a circle**

8 distinct people: total = 7!. Bad = glue B1,G1 -> 7 circular units -> 6! outer arrangements x 2 internal orders.

MISCONCEPTION **SUBTRACT 2 x 7!**

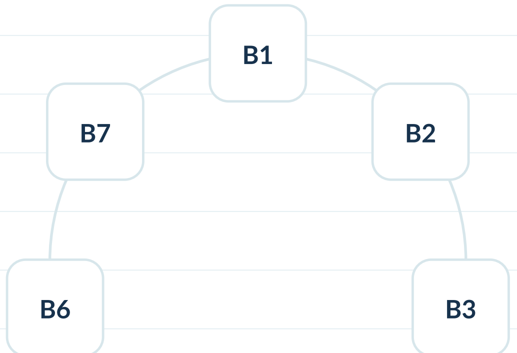
Tempting thought: “Treat the pair as a block but keep the same circular object count.”

Why it fails: Gluing reduces eight people to seven circular units.

Repair: Count units after gluing, then apply (units-1)!.

C6

Anchor group first



CIRCULAR GAP COUNT

n gaps

Arrange n separator objects around the circle. There is one gap after each anchor, so exactly n cyclic gaps. There are no two extra “end gaps” because the row has no ends.

Place the restricted group

MODEL

Choose gaps then assign

With 7 boys already arranged circularly, choose 5 of the 7 cyclic gaps and arrange 5 distinct girls in them: C(7,5) x 5!.

6! x C(7,5) x 5!

MISCONCEPTION

USE n+1 GAPS

Tempting thought: “Seven boys create eight gaps, like a line.”

Why it fails: A circle has no left and right end gaps; the last boy is adjacent back to the first.

Repair: Draw the cyclic wrap-around edge explicitly.

C7

Equal groups

MODEL **Gap capacity is tight**

n boys and n girls alternate around a round table. Arrange the n boys circularly: $(n-1)!$. Their n gaps must all be occupied by girls; arrange the girls in those n gaps: $n!$.

$$(n-1)! \times n!$$

VARIATION **$C(n,m) \times m!$**

If the restricted group is smaller, choose which cyclic gaps are used before arranging that group.

Why not 2 x ...?

MISCONCEPTION **TWO STARTING GENDERS**

Tempting thought: “Alternation can start with a boy or a girl, so multiply by 2.”

Why it fails: After circular rotation, those two apparent starts may represent the same seating class once one person/group is anchored.

Repair: Anchor one distinct person or one group first; do not double-count a mere rotation.

JEE TRIGGER **Circular gap engine**

No two girls / alternate / one group separated by the other -> arrange the larger separator group circularly first, then use its cyclic gaps.

C8 | LABELLED CHAIRS CHANGE EVERYTHING

A circle drawing does not guarantee circular equivalence

C8

Round table, unlabeled seats

UNLABELLED

Circular permutation

Only relative neighbour order matters -> rotations are identical -> (n-1)!.

Circular row of coloured / numbered chairs

LABELLED

Back to ST01

Seat identities distinguish rotations. Assign people to labelled chairs as ordinary labelled positions -> n!.

Misconception

MISCONCEPTION

CIRCLE DRAWN => (n-1)!

Tempting thought: “The seats form a circle, therefore divide by n.”

Why it fails: If chairs are labelled or otherwise distinguishable, a global rotation changes which person occupies which labelled position.

Repair: Ask whether positions or only relative neighbours define the outcome.

REAL-LIFE CHECK

Surface picture vs outcome definition

A wedding table with unnamed chairs: rotations may be equivalent. A carousel with numbered horses: rotations change the rider-horse assignment.

JEE MODELS: CIRCLE FIRST, RESTRICTIONS SECOND

Then import block / complement / gap

MODEL

2023 | NO TWO GIRLS C6

Arrange 7 boys circularly -> 6!. Their 7 cyclic gaps receive 5 distinct girls -> choose 5 gaps and arrange girls.

AIEEE 2003 | WOMEN SEPARATED C6/C7

Arrange 6 men circularly -> 5!. Six cyclic gaps appear. Choose 5 of 6 gaps and arrange 5 women -> 6 x 5!.

2017 | SPECIFIED PAIR NOT ADJACENT C5

Total 7! circular seatings of 8 people. Subtract pair-together count 2 x 6!.

METHOD ORDER Do not linearize too early

1. Decide symmetry / anchor.
2. Build the circular base.
3. Only then import block, complement or gap logic.

MISCONCEPTION CLINIC I

Rotation and reflection errors

M1

MISCONCEPTION

CIRCULAR_ROTATION_OVERCOUNT

Tempting thought: “Arrange n people in n! ways.”

Why it fails: Every circular arrangement appears n times through different starting points.

Repair: Fix one distinct person or divide the linear count by n.

MISCONCEPTION

REFLECTION DIVIDED AUTOMATICALLY

Tempting thought: “Clockwise reversal is the same because it is a circle.”

Why it fails: Ordinary seating preserves left/right neighbour positions; mirror images are generally different.

Repair: Only divide for reflection when the problem explicitly identifies flipped configurations.

MISCONCEPTION

LABELLED SEATS IGNORED

Tempting thought: “Round table always means (n-1)!.”

Why it fails: Labelled positions make rotations distinguishable outcomes.

Repair: Identify the outcome: person-to-seat assignment or relative neighbour pattern?

SELF-CHECK

What transformations preserve the outcome?

State the symmetry group of the problem in words before writing any factorial.

MISCONCEPTION CLINIC II

Block and gap transfer errors

M2

MISCONCEPTION

LINEAR BLOCK COUNT IN A CIRCLE

Tempting thought: “After gluing, arrange the units in (units)! ways.”

Why it fails: The reduced units are still circular, so rotations are still equivalent.

Repair: Use (number of circular units - 1)! outside the block.

MISCONCEPTION

GAP_OFF_BY_ONE

Tempting thought: “n circular anchors create n+1 gaps.”

Why it fails: There are no ends in a circle. The final anchor connects back to the first.

Repair: Count one cyclic gap after each anchor: exactly n.

MISCONCEPTION

ANCHOR RESTRICTION BROKEN

Tempting thought: “Fix any person even when that person belongs to a special block and then use a linear formula carelessly.”

Why it fails: Anchoring is safe, but the remaining restriction must still be represented correctly around the wrap-around edge.

Repair: After fixing the anchor, draw both neighbours and the closing edge explicitly.

REPAIR HABIT

Three-stage control

Always write circle base -> restriction engine -> internal arrangements in that order.

GUIDED PRACTICE + TRANSFER

Use symmetry language before arithmetic

GP

A. Complete

1 6 friends at an unlabelled round table

Base = ____!

2 8 people round table; A and B together

Outer circular units = ____; internal = ____

3 5 boys circularly fixed; 4 girls no two together

Cyclic gaps = ____; choose ____ of them

EXPLAIN **Topological difference**

Why do n anchors create n gaps in a circle but n+1 gaps in a line?

B. ExamSIDE fingerprints

2023	**5 girls, 7 boys; no two girls**	C6
2017	**B1 and G1 not adjacent**	C5
AIEEE 2003	**6 men, 5 women; no two women**	C6/C7

TRANSFER **Rebuild the picture**

A linear gap problem moved onto a round table is not “the same with one factorial changed.” The number of safe gaps itself changes because the endpoints disappear.

MASTERY + LOCAL COVERAGE AUDIT

Can you identify the symmetry before importing restrictions?

MASTER

You should now be able to...

1. explain why rotations are equivalent
2. derive (n-1)! by anchoring
3. avoid automatic reflection division
4. count circular blocks
5. use complement for a forbidden pair
6. use n cyclic gaps for group separation
7. handle alternation around a circle
8. recognize labelled-seat exceptions

Frozen ST09 ExamSIDE slice

REQUIRED **3 questions**

2023 no-two-girls round table
2017 specified boy/girl non-adjacent
AIEEE 2003 6 men/5 women separated

WHY ONLY 3? **Corpus discipline**

The audited JEE Main/AIEEE P&C; page exposes only a small direct circular-seating slice. We do not add unrelated circle geometry or non-JEE questions to inflate coverage.

OPEN EXAMSIDE CHAPTER

DERANGEMENTS & FORBIDDEN POSITIONS

When every object has a forbidden home

QUESTION

Essential Question

How do we count permutations when person i is forbidden from position i - and those forbidden events overlap heavily?

SCOPE

This subtopic builds

1. Fixed points vs derangements
2. Inclusion-exclusion formula for D_n
3. Recurrence
4. Exactly r fixed points
5. Partial derangement hybrids

OWNER

Boundary

A simple pair “not together” belongs to ST05. Here the forbidden condition is object-specific: each object has its own forbidden position.

C1 | START WITH A COAT CHECK

A wrong coat is a position-specific restriction

C1

REAL LIFE

Real life

Four guests hand in four labelled coats. At collection, nobody is allowed to receive their own coat. Each guest has exactly one forbidden coat.

HELPER

Visual model

Guest A cannot take coat A; B cannot take B; etc. Draw a 4x4 assignment grid and cross out the diagonal. A valid arrangement chooses exactly one cell in every row and column while avoiding all diagonal cells.

forbidden diagonal -> derangement

Meaning before notation

C2 | FIXED POINTS VS DERANGEMENTS

Name the event before choosing a formula

C2

TERM **Fixed point**

A fixed point is an object that lands in its original position: student i sits in seat i, card i goes to envelope i, etc.

TERM **Derangement**

A derangement is a permutation with zero fixed points. The count is written D_n or $!n$.

$$D_1=0,\quad D_2=1,\quad D_3=2,\quad D_4=9$$

Small values are useful for checks

C3 | WHY n! - n DOES NOT WORK

The bad events overlap

C3

MISCONCEPTION **Tempting subtraction**

“There are n! permutations. Remove n arrangements where somebody is fixed.” This treats each person-fixed event as one arrangement, which is false. Each event contains (n-1)! permutations, and intersections contain many permutations too.

REPAIR **Repair**

Let A_i = “object i is fixed.” We want permutations outside A_1 union ... union A_n. This is exactly an inclusion-exclusion problem.

$$n! - C(n,1)(n-1)! + C(n,2)(n-2)! - \dots$$

Alternating corrections repair the overlaps

C4

$$D_n = n![1 - 1/1! + 1/2! - 1/3! + \dots + (-1)^n/n!]$$

Subfactorial / derangement count

MEANING **What the kth term means**

Choose k objects that are forced to be fixed, then permute the remaining n-k objects. Inclusion-exclusion alternates signs because those fixed-point events overlap.

CHECK **Fast numerical check**

D_n is the nearest integer to $n!/e$ for moderate n. Use this as a plausibility check, not as the primary exact derivation.

C5 | RECURRENCE: BUILD D_n FROM SMALLER CASES

Condition on where object 1 goes

C5

$$D_n = (n-1)(D_{n-1} + D_{n-2})$$

Useful for hand computation and structural reasoning

CASE

Case A

Object 1 goes to position j and object j goes to position 1. Those two form a 2-cycle; the remaining n-2 objects must be deranged: D_{n-2}.

CASE

Case B

Object 1 goes to position j but j does not return to position 1. Relabel the remaining problem to obtain D_{n-1}. There are n-1 choices for j.

C6 | EXACTLY r FIXED POINTS

Choose the fixed objects, derange the rest

C6

$$\text{exactly } r \text{ fixed} = C(n,r)\, D_{n-r}$$

A choose-then-derange hybrid

MODEL

Why this works

First choose which r objects are correct. Every remaining object must be wrong - not merely “free.” That remaining subsystem is a derangement.

CONNECT

Related counts

At least one fixed = $n! - D_n$.

Exactly one fixed = $n\, D_{n-1}$.

C7 | PARTIAL DERANGEMENTS

One forced mapping changes the forbidden grid

C7

MODEL **Example structure**

Suppose card 1 is forced into envelope 2, while no card may enter its same-numbered envelope. Remove the used card and envelope first. Then inspect which diagonal restrictions remain.

H2 **Concept helper**

After removing card 1 and envelope 2, card 2 has no remaining “own envelope” restriction because envelope 2 is gone. Cards 3,... retain theirs. This is not simply D_5.

TRIGGER **JEE trigger**

Phrases such as “none in allotted seat”, “no card in matching envelope”, “wrong box”, “exactly r get their own” signal object-specific forbidden positions.

C8 | ASSIGNMENT-MATRIX VIEW

A diagonal of forbidden cells makes the structure visible

C8

Position ->	1	2	3	4
Object 1	X	OK	OK	OK
Object 2	OK	X	OK	OK
Object 3	OK	OK	X	OK
Object 4	OK	OK	OK	X

REPRESENT

Read the picture

A permutation corresponds to selecting exactly one allowed cell in each row and each column. Derangements are permutation matrices with zero diagonal entries.

NOTICE

Why this helps

Partial constraints are easier to audit visually before writing inclusion-exclusion.

JEE MODEL | FIVE STUDENTS, NO ALLOTTED SEAT

Direct derangement owner in the JEE Main corpus

MODEL

STEP 1

Translate

5 students + 5 labelled seats + nobody in own seat -> D_5. The wording already defines fixed points.

STEP 2

Compute

Use recurrence: $D_5 = 4(D_4 + D_3) = 4(9+2) = 44$. Or evaluate the inclusion-exclusion formula.

CHECK

Teacher check

If a proposed answer is near $5!/e = 44.15$, it is plausible. A count such as 4^5 is impossible because it ignores the one-to-one seat assignment.

MISCONCEPTION CLINIC I

Wrong-looking shortcuts are attractive for specific reasons

M1

MISCONCEPTION **n! - n**

Tempting: “remove one bad arrangement per person.”
Why wrong: each person-fixed event contains (n-1)! permutations.
Repair: define events and use inclusion-exclusion.

MISCONCEPTION **(n-1)^n**

Tempting: each object has n-1 wrong positions.
Why wrong: positions cannot be reused.
Repair: preserve the permutation/bijection constraint.

MISCONCEPTION **Subtract all fixed-point events once**

Tempting: n! - n(n-1)!.
Why wrong: permutations with two fixed points were subtracted twice, etc.
Repair: alternating intersections are essential.

MISCONCEPTION CLINIC II

Partial constraints must be re-audited after forced assignments

M2

MISCONCEPTION **Forced mapping -> D_{n-1}?**

Not automatically. Removing one card and one envelope can delete the forbidden position of a different card. Re-draw the remaining assignment grid.

MISCONCEPTION **Exactly r fixed -> $C(n,r)(n-r)!$**

Wrong because the remaining objects are allowed to create extra fixed points. The correct second factor is D_{n-r}.

REPAIR **Repair habit**

After any forced correct/wrong placement, list which object-position prohibitions survive before counting the residual problem.

GUIDED PRACTICE

Fade the scaffold: identify -> set up -> compute

GUIDED

TRY

A | Identify the owner

1. 6 letters into 6 addressed envelopes, none correctly addressed.
2. 7 students, exactly 2 in their allotted seats.
3. 5 cards, at least one card in its matching envelope.

SCAFFOLD

B | Complete the structure

1. $D_6 = 5(D_5 + \_\_\_)$.
2. Exactly 2 fixed among 7 = $C(7,2) \times \_\_\_$.
3. At least one fixed among 5 = $5! - \_\_\_$.

TRANSFER MAP

Recognize derangements even when the nouns change

TRANSFER

CONTEXT **Seats**

person i -> seat i forbidden

CONTEXT **Envelopes**

card i -> envelope i forbidden

CONTEXT **Coat check**

guest i -> coat i forbidden

NOTICE **Invariant**

The nouns change. The mathematics is a bijection with a forbidden diagonal. That is the recognition target.

MASTERY + LOCAL COVERAGE AUDIT

Release only when concept and corpus checks both pass

AUDIT

MASTERY **You should now be able to...**

1. Define fixed point and derangement.
2. Derive D_n by inclusion-exclusion.
3. Use the recurrence.
4. Count exactly r fixed points.
5. Handle a forced mapping by redrawing the residual forbidden grid.

COVERAGE **ExamSIDE owner audit**

JEE Main direct owner frozen: 2023 five allotted seats -> 44.

Legacy/JEE Advanced derangement variants are useful extensions but do not inflate JEE Main coverage.

BOUNDARY **Stop point**

General rook-polynomial machinery is beyond this subtopic. Use inclusion-exclusion only as far as the JEE-style forbidden grid requires.

FORBIDDEN STRINGS & INCLUSION-EXCLUSION

When several bad patterns can occur and overlap

QUESTION

Essential Question

How do we count permutations that must avoid one or more consecutive strings without double-counting the overlap between bad patterns?

SCOPE

This subtopic builds

1. String-as-block counting
2. Bad-event sets
3. Inclusion-exclusion
4. Compatible / incompatible overlaps
5. At least one vs neither
6. Pattern-avoidance recurrence

OWNER

Boundary

One group together is ST04. One group not together is ST05. ST11 begins when multiple forbidden/required patterns interact or when a local string can occur in more than one window.

C1 | A STRING IS A RIGID LOCAL ORDER

Glue the exact sequence, not merely its members

C1

REAL LIFE

Real life

A security code must not contain the exact substring 153. The digits 1,5,3 appearing somewhere is not enough; they must occur consecutively in that order.

MODEL

Block model

For permutations of 1,...,7 containing 153, glue [153] as one rigid block. Together with 2,4,6,7 there are 5 units, hence 5! arrangements.

contain exact string 153 -> [153] + other units

Order inside the string is fixed

C2 | TWO BAD STRINGS CREATE TWO EVENTS

Name A and B before subtracting

C2

EVENT **Event A**

A = permutations containing string 153.

Count A by gluing [153].

EVENT **Event B**

B = permutations containing string 2467.

Count B by gluing [2467].

$$\text{GOOD} = \text{TOTAL} - |A| - |B| + |A \cap B|$$

The overlap must be restored

C3 | COUNT THE INTERSECTION STRUCTURALLY

Can both strings occur at the same time?

C3

CASE

Disjoint strings

If the strings use disjoint symbols, both can appear as two separate rigid blocks. Count the two blocks plus all remaining singletons.

CASE

Overlapping strings

If the strings share symbols, first ask whether their overlap is compatible. Example: ABC and BCD can merge to ABCD; ABC and BAC cannot occupy the same copy of A,B,C consistently.

VISUAL

H2 helper

Draw the strings on transparent strips and slide them. Every compatible alignment corresponds to a merged super-string that can be counted as one block.

C4 | JEE MODEL: AVOID 153 AND 2467

A clean two-event inclusion-exclusion problem

C4

STEP 1 Bad counts

Total = 7!
|A| = 5! for [153]
|B| = 4! for [2467]

STEP 2 Overlap

Both strings use disjoint digits and together cover all seven digits. Treat [153] and [2467] as two units: |A intersection B| = 2!.

7! - 5! - 4! + 2! = 4898

JEE Main 2023

C5 | AT LEAST ONE PATTERN VS NEITHER

Choose the easier side of the complement

C5

FORM **At least one of A,B**

Count |A union B| = |A| + |B| - |A intersection B|.

FORM **Neither A nor B**

Count Total - |A union B|. This is often cleaner than trying to construct all valid permutations directly.

TRIGGER **Decision cue**

If the valid condition says “avoid”, “neither”, or “does not contain”, ask whether the forbidden side has a simpler block description.

C6 | WHY "WINDOWS x ARRANGEMENTS" CAN OVERCOUNT

A sequence may contain the target pattern more than once

C6

MISCONCEPTION

Tempting method

Choose a starting window for the required string, choose its orientation, then arrange the rest.

WHY

Failure mode

If a final sequence contains the target string in two windows, it gets counted twice. Repeated symbols make this especially dangerous.

REPAIR

Repair

Use event sets for windows and inclusion-exclusion, or a recurrence/state method. Before multiplying by "number of starting positions," prove each valid object has exactly one marked occurrence.

C7 | PATTERN AVOIDANCE BY STATE

Local restrictions can create Fibonacci-type recurrences

C7

MODEL

Binary example

Count n-digit binary strings with no consecutive 0s. Split by last digit: ending in 1 can follow any valid shorter string; ending in 0 must follow a string ending in 1.

$$a_n = a_{n-1} + a_{n-2}$$

A two-state recurrence

CONNECT

Why it belongs here

The forbidden local pattern is “00.” When block/complement counting is awkward, a state that remembers the suffix can be cleaner.

C8 | OVERLAP AUDIT

Before inclusion-exclusion, classify intersections

C8

CLASSIFY

Three intersection types

1. Disjoint: two independent rigid blocks.
2. Compatible overlap: merge into one longer super-string.
3. Incompatible overlap: intersection may be empty or require separate copies if repetitions exist.

AUDIT

Three questions

1. Do the strings share symbols?
2. Can shared symbols align consistently?
3. Can the underlying multiset supply multiple copies if needed?

MISCONCEPTION CLINIC I

The intersection is where most errors hide

M1

MISCONCEPTION **Subtract A and B only**

Fails because arrangements containing both strings are removed twice. Repair with +|A intersection B|.

MISCONCEPTION **Multiply start positions blindly**

Fails when one object can contain several marked occurrences. Repair by checking uniqueness of the marked window.

MISCONCEPTION **Treat “members together” as “exact string”**

[153] means exactly 1-5-3. A block of the set {1,5,3} would have 3! internal orders and is a different event.

MISCONCEPTION CLINIC II

At least once is an event union, not a single block

M2

MISCONCEPTION **"At least once" -> one chosen occurrence**

A valid sequence may contain multiple occurrences, so marking one can duplicate the same final sequence.

MISCONCEPTION **Assume overlap impossible because strings differ**

Different strings can share a suffix/prefix. Example ABC and BCD overlap as ABCD. Always test alignments.

REPAIR **Repair habit**

Draw the forbidden strings, list event intersections, then decide block / complement / recurrence. Do not start with arithmetic.

GUIDED PRACTICE

Build the event structure before evaluating

GUIDED

TRY

A | Event setup

1. Permutations of 1,...,6 avoiding 123.
2. Permutations of 1,...,7 avoiding both 123 and 456.
3. Binary strings length 8 with no 00.

REASON

B | Intersection audit

1. For strings 123 and 345, can both occur?
2. For ABC and BCD, what merged string represents overlap?
3. For repeated letters, when can two copies of the same string occur?

SOURCE QUALITY CHECK

Do not inherit a counting shortcut without verifying uniqueness

CHECK

AUDIT NOTE

ExamSIDE 2023 AP-window item - deferred from the required set

The chapter contains a nine-digit repeated-symbol problem requiring an AP triple “at least once.” Its displayed explanation multiplies by the seven starting windows and two orientations. Because repeated symbols can create more than one qualifying window in the same final sequence, that shortcut needs an overlap audit before being used pedagogically. This guide therefore does not freeze it as a clean ST11 owner.

SOURCE GATE

Principle

Coverage completeness does not override mathematical integrity. A questionable source solution is flagged and deferred, not silently propagated.

TRANSFER MAP

One idea, several surface forms

TRANSFER

CONTEXT **Digit permutations**

avoid exact substrings

CONTEXT **Words**

forbid letter patterns

CONTEXT **Binary strings**

avoid 00 / 11 patterns

NOTICE **Invariant**

Define bad events by local patterns, audit intersections, then choose block + inclusion-exclusion or a suffix-state recurrence.

MASTERY + LOCAL COVERAGE AUDIT

Keep the owner set strict

AUDIT

MASTERY **You should now be able to...**

1. Count an exact required/forbidden string as a rigid block.
2. Use two-event inclusion-exclusion.
3. Classify string intersections.
4. Detect window-marking overcount.
5. Use a recurrence for local pattern avoidance.

COVERAGE **ExamSIDE owner audit**

Direct JEE Main owner frozen: 2023 permutations avoiding 153 and 2467 -> 4898.

The AP “at least once” item is deferred pending overlap-correct solution review.

BOUNDARY **Stop point**

General automata / Goulden-Jackson methods are beyond JEE Main scope; use only the state depth needed by the forbidden suffix.

HYBRID PERMUTATION STRUCTURES

Choose, classify, assign, then arrange - in the correct order

QUESTION

Essential Question

When a JEE problem mixes selection, grouping and ordering, how do we identify which decisions are unordered and which are labelled/ordered?

SCOPE

This subtopic builds

1. Choose -> arrange
2. Choose group -> assign labels
3. Multiplicity-pattern cases
4. Distribution + permutation
5. Avoiding accidental combination/permutation swaps

OWNER

Boundary

This is the capstone for permutation-enabled hybrids. Problems whose primary engine is stars-and-bars, committee selection, or pure nCr identities remain outside the permutation book.

C1 | DECISION PIPELINE

Do not compress unlike decisions into one formula

C1

CLASSIFY -> CHOOSE -> ASSIGN -> ARRANGE

Some stages may disappear in a specific problem

STAGE **Unlabelled choice**

Choosing 2 vowels from 5 is a combination stage: the pair {A,E} has no internal order yet.

STAGE **Labelled placement**

Once four distinct letters are selected, placing them into four ordered positions contributes 4!.

C2 | CHOOSE THEN ARRANGE

The most common hybrid

C2

MODEL **Word model**

Need exactly 2 vowels and 2 consonants. First choose which 2 vowel symbols and which 2 consonant symbols. Then arrange the 4 chosen distinct letters.

$$C(V,2)\ C(C,2)\ 4!$$

Selection x permutation

TRAP **Misconception**

Using $P(V,2)P(C,2)$ and then $4!$ double-counts the internal order of the selected vowels/consonants.

C3 | CHOOSE A GROUP, THEN ASSIGN LABELLED DESTINATIONS

Group identity can make floor choices ordered

C3

STEP 1 **Lift model**

Choose which 4 of 9 people form the size-4 exit group. The size-5 group is then determined.

STEP 2 **Assign floors**

If there are 8 permitted floors and the two groups must exit at different floors, the floor assignment is 8P2, because “floor for group of 4” and “floor for group of 5” are labelled roles.

$$C(9,4) \times 8P2 = 7056$$

Choose people, then assign destinations

C4 | MULTIPLICITY PATTERNS FIRST

When symbols may repeat, classify count patterns before arranging

C4

MODEL

Six-letter words from MATHS

Every used letter must appear at least twice. The number of distinct symbols used can only be 1, 2, or 3. That case split is the structural first step.

PATTERN

Patterns

k=1: (6)
k=2: (2,4), (3,3), (4,2)
k=3: (2,2,2)

H2

Concept helper

Separate which symbols are used from how many times each is used from where those copies are placed. Mixing these three layers causes most hybrid errors.

C5 | OWNER TEST: WHAT IS THE PRIMARY ENGINE?

Hybrid does not mean every P&C problem belongs here

C5

INCLUDE **Include in permutation book**

1. Choose letters then arrange them.
2. Choose people then assign labelled roles/floors.
3. Choose multiplicity pattern then count word arrangements.
4. Distribution where final recipient labels create ordered assignments.

EXCLUDE **Exclude / defer**

1. Pure committee selection.
2. Stars-and-bars with identical objects only.
3. nCr identities with no ordering stage.
4. Geometry subset counting.

C6 | LABELLED VS UNLABELLED GROUPS

A tiny wording change can multiply the answer

C6

MODEL

Unlabelled groups

Split 9 people into groups of sizes 4 and 5. Since the sizes differ, choosing the 4-person group determines the split: C(9,4).

CONNECT

Labelled destinations

Now send the two groups to floors F and G. The roles “group at F” and “group at G” are distinct; assignment adds an ordered stage.

HELPER

Swap test

If swapping two chosen groups/roles produces a different outcome, the assignment stage is ordered.

C7 | POSITION-SLOT VIEW FOR HYBRIDS

After selection, return to slots

C7

SLOTS

Four selected letters

Once {A,E,N,R} is fixed, the final word is an ordered filling of four slots. Selection is finished; the remaining count is 4!.

SLOTS

Repeated copies

If counts are (2,2,2), six slots contain three pairs of identical symbols. Use 6!/(2!2!2!) after choosing the three symbols.

choose the ingredients x arrange the selected multiset

Two different kinds of symmetry

JEE MODEL I | INCONSEQUENTIAL

Distinct symbol inventory before choosing

MODEL

STEP 1

Inventory

Distinct vowels: I,O,E,U,A -> 5.
Distinct consonants: N,C,S,Q,T,L -> 6. The original word has repeated letters, but the question forbids repeating a letter in the 4-letter word.

STEP 2

Count

Choose 2 vowels and 2 consonants, then arrange all four distinct chosen letters: $C(5,2)C(6,2)4! = 3600$.

TRAP

Teacher check

Do not treat repeated occurrences in the source word as separate available symbols when the formed word itself cannot repeat a letter.

JEE MODEL II | MATHS WITH “USED => AT LEAST TWICE”

Case structure controls the permutation count

MODEL

STEP 1 Cases

One symbol: 5 words.
Two symbols: choose pair, then counts 2+4 or 3+3 or 4+2.
Three symbols: choose 3, then pattern 2+2+2.

STEP 2 Evaluate

k=2 contributes C(5,2)[15+20+15] = 500.
k=3 contributes C(5,3) x 6!/(2!^3) = 900.
Total = 5+500+900 = 1405.

H2 Concept helper

The allowed multiplicity partitions of 6 are the backbone. Only after freezing a partition do you choose symbols and arrange copies.

MISCONCEPTION CLINIC I

Selection and arrangement are different symmetries

M1

MISCONCEPTION **Use P instead of C during selection**

Double-counts because the final 4! already orders the chosen letters. Repair: unordered ingredient selection first.

MISCONCEPTION **Choose two floors with C(8,2) only**

Misses which floor belongs to which differently-sized group. Repair: use 8P2 or C(8,2)x2!.

MISCONCEPTION **Ignore multiplicity patterns**

Counting 5^6 and filtering later is possible in principle but opaque and error-prone. Repair: classify allowed count vectors first.

MISCONCEPTION CLINIC II

The source inventory and the output restriction are different

M2

MISCONCEPTION

Repeated letter in source word -> multiple copies available

Not necessarily. If the problem says “without repeating any letter,” use distinct symbols, even if the source word contains repeated occurrences.

MISCONCEPTION

Every hybrid needs a combination stage

False. If roles are assigned directly to labelled slots, slot filling may count selection and order simultaneously. Use the representation that exposes restrictions best.

REPAIR

Repair habit

Write the decision pipeline in words before symbols: “choose X, then assign Y, then arrange Z.” Cross out any stage that is already determined.

GUIDED PRACTICE

Annotate each factor with its decision

GUIDED

TRY

A | Factor labels

1. C(7,3) x 3! : what does each factor decide?
2. C(9,4) x 8P2 : identify group-choice vs destination assignment.
3. C(5,3) x 6!/(2!^3) : identify symbol-choice vs multiset arrangement.

CLASSIFY

B | Owner decisions

1. 2 vowels + 2 consonants from a word, no repetition.
2. 8 people in 3 labelled cars with capacities.
3. Choose a 5-person committee only. Which one is outside this book?

TRANSFER MAP

Hybrid questions are pipelines, not formulas

TRANSFER

CONTEXT **Words**

choose categories -> arrange

CONTEXT **Lifts / cars**

choose groups -> assign destinations

CONTEXT **Repeated symbols**

choose count pattern -> multiset arrange

NOTICE **Invariant**

Every factor in the final product should correspond to one explicit decision. If you cannot name the decision, the factor is suspect.

MASTERY + LOCAL COVERAGE AUDIT

Capstone owner rules

AUDIT

MASTERY

You should now be able to...

1. Separate unordered and ordered stages.
2. Use choose-then-arrange correctly.
3. Detect labelled destination assignments.
4. Classify multiplicity patterns before arranging.
5. Reject pure-combination problems from a permutation-only resource.

COVERAGE

ExamSIDE owner audit

Frozen JEE Main owners include: 2026 INCONSEQUENTIAL; 2026 lift groups to distinct floors; 2025 MATHS multiplicity condition; 2023 UNIVERSE.

These are permutation-enabled hybrids, not pure selection questions.

BOUNDARY

Stop point

Pure distributions, stars-and-bars, committees, and combination identities are explicitly excluded from the permutation-only book.